

Functional Constraints

Flame competition constraint within pocket regions.

Penalty coefficient 0.0100

Maximum allowable heat flux density ratio (max/min) 1000.00

The linear burning rate of the first conditional fuel is greater than that of the second.

Penalty coefficient 0.0100

Constraint on maximum kinetic flame heat flux density.

Penalty coefficient 0.0100

Maximum kinetic flame heat flux density in inter-pocket medium 1.00e+09

Maximum kinetic flame heat flux density in skeleton structure 1.00e+08

Maximum kinetic flame heat flux density in pocket region outside skeleton 1.00e+08

Constraint on skeleton thickness to pore diameter ratio.

Penalty coefficient 0.0100

Threshold for skeleton thickness to pore diameter ratio 3.00

Constraint on skeleton thickness to large oxidizer particle diameter ratio.

Penalty coefficient 0.0100

Threshold for skeleton thickness to large oxidizer particle diameter ratio 1.00

Parametric Constraints

Constraint on allowable surface temperature range

Minimum surface temperature 600 K

Maximum surface temperature 900 K

Differential Evolution Algorithm

Population size: 480

Mutation force: 0.5

Crossover probability: 0.9

Number of threads: 32

Termination strategy: OrTerminationStrategy

Group Combustion Solver Parameters

Shared Parameters (All Compositions)

Pre-exponential coefficient for gas phase reactions in inter-pocket medium 5357005.305021299

Parameter bounds [1.000E+005; 1.000E+013]

Activation energy for gas phase reactions in inter-pocket medium 50000.65656897834

Parameter bounds [5.000E+004; 2.500E+005]

Pre-exponential factor for gas-phase reactions in pocket region outside skeleton 1061408.0865843806

Parameter bounds [1.000E+005; 1.000E+013]

Activation energy for gas-phase reactions in pocket region outside skeleton 50000.000000169195

Parameter bounds [5.000E+004; 2.500E+005]

Reaction order for gas-phase chemistry in inter-pocket medium 2.34057781398687

Parameter bounds [0.000E+000; 2.500E+000]

Reaction order for gas-phase chemistry outside skeleton structure 0.6366005763584496

Parameter bounds [0.000E+000; 2.500E+000]

Enthalpy difference between vaporization and binder decomposition 664321.1529952802

Parameter bounds [-1.000E+007; 1.000E+007]

Diffusion flame height correlation coefficient 1.7279

Parameter bounds [1.000E-003; 1.000E+001]

Power law exponent for burning rate dependency on A (skeleton thickness) 1.0577

Parameter bounds [0.000E+000; 3.000E+000]

Power law exponent for burning rate dependency on B (pore diameter) 1.0577

Parameter bounds [0.000E+000; 3.000E+000]

Radiation temperature correlation coefficient (average layer temperature) 0.0000

Parameter bounds [0.000E+000; 1.000E+000]

Diffusion-flame standoff pressure-factor C_{rxn} 0.00218273

Parameter bounds [0.000E+000; 5.000E+001]

Diffusion-flame standoff size-exponent m 7.06268E-09

Parameter bounds [0.000E+000; 2.000E+000]

Bimodal-packing heat-feedback K_{pack} 10

Parameter bounds [0.000E+000; 1.000E+001]
Diffusion-flame standoff pressure-exponent n_p 4
Parameter bounds [2.000E+000; 4.000E+000]
WSB condensed-phase reaction K_{wsb} 0.260555
Parameter bounds [0.000E+000; 5.000E+000]
AP premixed-flame conductance K_{AP} 4154.43
Parameter bounds [0.000E+000; 5.000E+003]
AP self-deflagration pressure-exponent n_{AP} 1.33656
Parameter bounds [7.700E-001; 2.500E+000]
Bimodal-packing pressure-decay exponent a_{pack} 0.218788
Parameter bounds [0.000E+000; 2.000E+000]

Bas_2 (includes Bas_3, Bas_4) - Condensed Phase Parameters

Pre-exponential coefficient for condensed phase reactions 9999089085895.756
Parameter bounds [1.000E+000; 1.000E+013]
Activation energy for condensed phase reactions 186913.85452062284
Parameter bounds [5.000E+004; 3.000E+005]
Pre-exponential factor for gas-phase reactions within skeleton structure 3189109.6216363707
Parameter bounds [1.000E+005; 1.000E+013]
Activation energy for gas-phase reactions within skeleton structure 50564.1758035431
Parameter bounds [5.000E+004; 2.500E+005]
Reaction order for gas-phase chemistry in skeleton structure 9.441103598720525E-09
Parameter bounds [0.000E+000; 2.500E+000]
Coefficient A for metal combustion heat flux density in skeleton structure, 2.611050178329435E-07
Parameter bounds [1.000E-010; 1.000E-003]
Coefficient B for metal combustion heat flux density in skeleton structure, 1.0063368948751587E-12
Parameter bounds [1.000E-012; 1.000E-005]

Bas_1 - Condensed Phase Parameters

Pre-exponential coefficient for condensed phase reactions 9999994087626.809
Parameter bounds [1.000E+000; 1.000E+013]
Activation energy for condensed phase reactions 154399.75165551863
Parameter bounds [5.000E+004; 3.000E+005]
Pre-exponential factor for gas-phase reactions within skeleton structure 32751495.684050556
Parameter bounds [1.000E+005; 1.000E+013]
Activation energy for gas-phase reactions within skeleton structure 50000.00067393954
Parameter bounds [5.000E+004; 2.500E+005]
Reaction order for gas-phase chemistry in skeleton structure 1.5609903382427865
Parameter bounds [0.000E+000; 2.500E+000]
Coefficient A for metal combustion heat flux density in skeleton structure, 7.67123688946657E-07
Parameter bounds [1.000E-010; 1.000E-003]
Coefficient B for metal combustion heat flux density in skeleton structure, 1.0006198628797108E-12
Parameter bounds [1.000E-012; 1.000E-005]

Bas_0 - Condensed Phase Parameters

Pre-exponential coefficient for condensed phase reactions 9999984802342.879
Parameter bounds [1.000E+000; 1.000E+013]
Activation energy for condensed phase reactions 148488.83546160304
Parameter bounds [5.000E+004; 3.000E+005]
Pre-exponential factor for gas-phase reactions within skeleton structure 120339758.59855561
Parameter bounds [1.000E+005; 1.000E+013]
Activation energy for gas-phase reactions within skeleton structure 50000.0001174551
Parameter bounds [5.000E+004; 2.500E+005]
Reaction order for gas-phase chemistry in skeleton structure 0.683077443334889
Parameter bounds [0.000E+000; 2.500E+000]
Coefficient A for metal combustion heat flux density in skeleton structure, 4.3386425379674314E-07
Parameter bounds [1.000E-010; 1.000E-003]
Coefficient B for metal combustion heat flux density in skeleton structure, 1.4462140796009575E-07
Parameter bounds [1.000E-012; 1.000E-005]

Optimization Results

Objective function value 0.0083

Total penalty from constraint violations 0.0000

Propellant Composition "Bas_2"

At pressure 1 MPa

Experimental burning rate 16.94 mm/s

Calculated burning rate 16.37 mm/s

First Conditional Fuel

Surface temperature 661.26 K

Linear burning rate 11.24 mm/s

Mass decomposition rate $18.65 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Kinetic flame heat flux density to surface $2.26\text{e}+07 \text{ W/m}^2$

Kinetic flame height 13.5 μm

Average kinetic flame density 1.82 kg/m^3

Average kinetic flame temperature 2,184.13 K

Binder sublimation heat flux density $2.26\text{e}+07 \text{ W/m}^2$

Surface heat balance error 0.0046 W/m^2

Secondary Conditional Fuel

Surface temperature 655.78 K

Linear burning rate 8.97 mm/s

Mass decomposition rate $14.88 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $420,755.22 \text{ W/m}^2$

Kinetic flame height 406.23 μm

Average kinetic flame density 2.63 kg/m^3

Average kinetic flame temperature 1,510.39 K

Skeleton Structure Region

Kinetic flame heat flux density to surface $8.48\text{e}+06 \text{ W/m}^2$

Kinetic flame height 11.03 μm

Average kinetic flame density 3.53 kg/m^3

Average kinetic flame temperature 1,123.39 K

Heat flux density due to metal combustion, $4.44\text{e}+06 \text{ W/m}^2$

Effective thermal conductivity $0.44 \text{ W/(m}\cdot\text{K)}$

Conductive thermal conductivity $0.4 \text{ W/(m}\cdot\text{K)}$

Radiative thermal conductivity $0.034 \text{ W/(m}\cdot\text{K)}$

Conductive thermal conductivity balance error $-4.092294297874943\text{E}-08$

Skeleton layer thickness $6.35\text{e}-05 \text{ m}$

Pore diameter $2.12\text{e}-05 \text{ m}$

Diffusion flame heat flux density $46,219.76 \text{ W/m}^2$

Diffusion flame height 6,378.71 μm

Heat flux density to surface within skeletal structure $4.64\text{e}+06 \text{ W/m}^2$

Heat flux density to surface outside skeletal structure $269,732.5 \text{ W/m}^2$

Total heat flux density to surface $1.62\text{e}+07 \text{ W/m}^2$

Binder sublimation heat flux density $1.62\text{e}+07 \text{ W/m}^2$

Surface heat balance error -0.0039 W/m^2

At pressure 4.06 MPa

Experimental burning rate 35.09 mm/s

Calculated burning rate 35.63 mm/s

First Conditional Fuel

Surface temperature 703.13 K

Linear burning rate 56.15 mm/s

Mass decomposition rate $93.15 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Kinetic flame heat flux density to surface $1.18\text{e}+08 \text{ W/m}^2$

Kinetic flame height 2.54 μm

Average kinetic flame density 7.3 kg/m³
Average kinetic flame temperature 2,205.07 K
Binder sublimation heat flux density 1.18e+08 W/m²
Surface heat balance error -0.019 W/m²

Secondary Conditional Fuel

Surface temperature 670.59 K
Linear burning rate 16.37 mm/s
Mass decomposition rate 27.16 kg·s⁻¹·m⁻²

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 566,468.77 W/m²
Kinetic flame height 299.12 μm
Average kinetic flame density 10.61 kg/m³
Average kinetic flame temperature 1,517.79 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 1.23e+07 W/m²
Kinetic flame height 7.5 μm
Average kinetic flame density 14.23 kg/m³
Average kinetic flame temperature 1,130.79 K
Heat flux density due to metal combustion, 7.9e+06 W/m²
Effective thermal conductivity 0.42 W/(m·K)
Conductive thermal conductivity 0.4 W/(m·K)
Radiative thermal conductivity 0.018 W/(m·K)
Conductive thermal conductivity balance error -4.092294297874943E-08
Skeleton layer thickness 3.36e-05 m
Pore diameter 1.12e-05 m
Diffusion flame heat flux density 154,276.59 W/m²
Diffusion flame height 1,901.4 μm
Heat flux density to surface within skeletal structure 6.66e+06 W/m²
Heat flux density to surface outside skeletal structure 379,337.92 W/m²
Total heat flux density to surface 2.09e+07 W/m²
Binder sublimation heat flux density 2.09e+07 W/m²
Surface heat balance error -0.0037 W/m²

At pressure 6.5 MPa

Experimental burning rate 44.85 mm/s
Calculated burning rate 42.03 mm/s

First Conditional Fuel

Surface temperature 718.44 K
Linear burning rate 96.46 mm/s
Mass decomposition rate 160.03 kg·s⁻¹·m⁻²
Kinetic flame heat flux density to surface 2.07e+08 W/m²
Kinetic flame height 1.44 μm
Average kinetic flame density 11.66 kg/m³
Average kinetic flame temperature 2,212.72 K
Binder sublimation heat flux density 2.07e+08 W/m²
Surface heat balance error -0.039 W/m²

Secondary Conditional Fuel

Surface temperature 673.8 K
Linear burning rate 18.58 mm/s
Mass decomposition rate 30.83 kg·s⁻¹·m⁻²

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 674,882.56 W/m²
Kinetic flame height 250.59 μm
Average kinetic flame density 16.98 kg/m³
Average kinetic flame temperature 1,519.4 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 1.5e+07 W/m²
Kinetic flame height 6.13 μm
Average kinetic flame density 22.78 kg/m³

Average kinetic flame temperature 1,132.4 K
 Heat flux density due to metal combustion, $8.94\text{e}+06 \text{ W/m}^2$
 Effective thermal conductivity $0.42 \text{ W/(m}\cdot\text{K)}$
 Conductive thermal conductivity $0.4 \text{ W/(m}\cdot\text{K)}$
 Radiative thermal conductivity $0.016 \text{ W/(m}\cdot\text{K)}$
 Conductive thermal conductivity balance error $-4.092294297874943\text{E}-08$
 Skeleton layer thickness $2.94\text{e}-05 \text{ m}$
 Pore diameter $9.8\text{e}-06 \text{ m}$
 Diffusion flame heat flux density $137,981.09 \text{ W/m}^2$
 Diffusion flame height $2,123.63 \mu\text{m}$
 Heat flux density to surface within skeletal structure $6.49\text{e}+06 \text{ W/m}^2$
 Heat flux density to surface outside skeletal structure $491,530.63 \text{ W/m}^2$
 Total heat flux density to surface $2.19\text{e}+07 \text{ W/m}^2$
 Binder sublimation heat flux density $2.19\text{e}+07 \text{ W/m}^2$
 Surface heat balance error -0.0039 W/m^2

Propellant Composition "Bas_3"

At pressure 1 MPa

Experimental burning rate 15.24 mm/s
 Calculated burning rate 15.01 mm/s

First Conditional Fuel

Surface temperature 661.26 K
 Linear burning rate 11.24 mm/s
 Mass decomposition rate $18.65 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
 Kinetic flame heat flux density to surface $2.26\text{e}+07 \text{ W/m}^2$
 Kinetic flame height $13.5 \mu\text{m}$
 Average kinetic flame density 1.82 kg/m^3
 Average kinetic flame temperature 2,184.13 K
 Binder sublimation heat flux density $2.26\text{e}+07 \text{ W/m}^2$
 Surface heat balance error 0.0046 W/m^2

Secondary Conditional Fuel

Surface temperature 661.26 K
 Linear burning rate 11.24 mm/s
 Mass decomposition rate $18.65 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $336,619.14 \text{ W/m}^2$
 Kinetic flame height $506.13 \mu\text{m}$
 Average kinetic flame density 2.62 kg/m^3
 Average kinetic flame temperature 1,513.13 K

Skeleton Structure Region

Kinetic flame heat flux density to surface $5.94\text{e}+07 \text{ W/m}^2$
 Kinetic flame height $3.3 \mu\text{m}$
 Average kinetic flame density 2.42 kg/m^3
 Average kinetic flame temperature 1,642.13 K
 Heat flux density due to metal combustion, $4.94\text{e}+06 \text{ W/m}^2$
 Effective thermal conductivity $0.39 \text{ W/(m}\cdot\text{K)}$
 Conductive thermal conductivity $0.22 \text{ W/(m}\cdot\text{K)}$
 Radiative thermal conductivity $0.17 \text{ W/(m}\cdot\text{K)}$
 Conductive thermal conductivity balance error $-5.5152085010057306\text{E}-08$
 Skeleton layer thickness $5\text{e}-05 \text{ m}$
 Pore diameter $1.67\text{e}-05 \text{ m}$
 Diffusion flame heat flux density $94,261.87 \text{ W/m}^2$
 Diffusion flame height $3,196.14 \mu\text{m}$
 Heat flux density to surface within skeletal structure $2.01\text{e}+07 \text{ W/m}^2$
 Heat flux density to surface outside skeletal structure $231,287.48 \text{ W/m}^2$
 Total heat flux density to surface $2.05\text{e}+07 \text{ W/m}^2$
 Binder sublimation heat flux density $2.05\text{e}+07 \text{ W/m}^2$
 Surface heat balance error -0.0068 W/m^2

At pressure 4.06 MPa

Experimental burning rate 37.34 mm/s
Calculated burning rate 38.11 mm/s

First Conditional Fuel

Surface temperature 703.13 K
Linear burning rate 56.15 mm/s
Mass decomposition rate $93.15 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
Kinetic flame heat flux density to surface $1.18\text{e}+08 \text{ W/m}^2$
Kinetic flame height 2.54 μm
Average kinetic flame density 7.3 kg/m^3
Average kinetic flame temperature 2,205.07 K
Binder sublimation heat flux density $1.18\text{e}+08 \text{ W/m}^2$
Surface heat balance error -0.019 W/m^2

Secondary Conditional Fuel

Surface temperature 676.71 K
Linear burning rate 20.83 mm/s
Mass decomposition rate $34.55 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $446,592.54 \text{ W/m}^2$
Kinetic flame height 378.04 μm
Average kinetic flame density 10.58 kg/m^3
Average kinetic flame temperature 1,520.85 K

Skeleton Structure Region

Kinetic flame heat flux density to surface $8.39\text{e}+07 \text{ W/m}^2$
Kinetic flame height 2.32 μm
Average kinetic flame density 9.76 kg/m^3
Average kinetic flame temperature 1,649.85 K
Heat flux density due to metal combustion, $7.36\text{e}+06 \text{ W/m}^2$
Effective thermal conductivity $0.31 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity $0.22 \text{ W/(m}\cdot\text{K)}$
Radiative thermal conductivity $0.086 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity balance error $-5.5152085010057306\text{E-}08$
Skeleton layer thickness $2.6\text{e-}05 \text{ m}$
Pore diameter $8.68\text{e-}06 \text{ m}$
Diffusion flame heat flux density $309,950.8 \text{ W/m}^2$
Diffusion flame height 967.02 μm
Heat flux density to surface within skeletal structure $2.13\text{e}+07 \text{ W/m}^2$
Heat flux density to surface outside skeletal structure $342,455.99 \text{ W/m}^2$
Total heat flux density to surface $2.67\text{e}+07 \text{ W/m}^2$
Binder sublimation heat flux density $2.67\text{e}+07 \text{ W/m}^2$
Surface heat balance error 0.0032 W/m^2

At pressure 6.5 MPa

Experimental burning rate 50.5 mm/s
Calculated burning rate 47.6 mm/s

First Conditional Fuel

Surface temperature 718.44 K
Linear burning rate 96.46 mm/s
Mass decomposition rate $160.03 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
Kinetic flame heat flux density to surface $2.07\text{e}+08 \text{ W/m}^2$
Kinetic flame height 1.44 μm
Average kinetic flame density 11.66 kg/m^3
Average kinetic flame temperature 2,212.72 K
Binder sublimation heat flux density $2.07\text{e}+08 \text{ W/m}^2$
Surface heat balance error -0.039 W/m^2

Secondary Conditional Fuel

Surface temperature 680.54 K
Linear burning rate 24.16 mm/s

Mass decomposition rate $40.08 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $520,765.07 \text{ W/m}^2$

Kinetic flame height $323.46 \text{ }\mu\text{m}$

Average kinetic flame density 16.94 kg/m^3

Average kinetic flame temperature $1,522.77 \text{ K}$

Skeleton Structure Region

Kinetic flame heat flux density to surface $1\text{e}+08 \text{ W/m}^2$

Kinetic flame height $1.94 \text{ }\mu\text{m}$

Average kinetic flame density 15.62 kg/m^3

Average kinetic flame temperature $1,651.77 \text{ K}$

Heat flux density due to metal combustion, $8.22\text{e}+06 \text{ W/m}^2$

Effective thermal conductivity $0.3 \text{ W/(m}\cdot\text{K)}$

Conductive thermal conductivity $0.22 \text{ W/(m}\cdot\text{K)}$

Radiative thermal conductivity $0.073 \text{ W/(m}\cdot\text{K)}$

Conductive thermal conductivity balance error $-5.5152085010057306\text{E}-08$

Skeleton layer thickness $2.23\text{e}-05 \text{ m}$

Pore diameter $7.42\text{e}-06 \text{ m}$

Diffusion flame heat flux density $271,197.91 \text{ W/m}^2$

Diffusion flame height $1,103.79 \text{ }\mu\text{m}$

Heat flux density to surface within skeletal structure $1.64\text{e}+07 \text{ W/m}^2$

Heat flux density to surface outside skeletal structure $441,959.24 \text{ W/m}^2$

Total heat flux density to surface $2.85\text{e}+07 \text{ W/m}^2$

Binder sublimation heat flux density $2.85\text{e}+07 \text{ W/m}^2$

Surface heat balance error -0.0073 W/m^2

Propellant Composition "Bas_4"

At pressure 1 MPa

Experimental burning rate 9.73 mm/s

Calculated burning rate 9.82 mm/s

First Conditional Fuel

Surface temperature 661.26 K

Linear burning rate 11.24 mm/s

Mass decomposition rate $18.65 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Kinetic flame heat flux density to surface $2.26\text{e}+07 \text{ W/m}^2$

Kinetic flame height $13.5 \text{ }\mu\text{m}$

Average kinetic flame density 1.82 kg/m^3

Average kinetic flame temperature $2,184.13 \text{ K}$

Binder sublimation heat flux density $2.26\text{e}+07 \text{ W/m}^2$

Surface heat balance error 0.0046 W/m^2

Secondary Conditional Fuel

Surface temperature 632.91 K

Linear burning rate 3.35 mm/s

Mass decomposition rate $5.56 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $15,994.54 \text{ W/m}^2$

Kinetic flame height $2,557.69 \text{ }\mu\text{m}$

Average kinetic flame density 4.74 kg/m^3

Average kinetic flame temperature 837.45 K

Skeleton Structure Region

Kinetic flame heat flux density to surface $5.75\text{e}+06 \text{ W/m}^2$

Kinetic flame height $10.57 \text{ }\mu\text{m}$

Average kinetic flame density 4.24 kg/m^3

Average kinetic flame temperature 936.95 K

Heat flux density due to metal combustion, $5.37\text{e}+06 \text{ W/m}^2$

Effective thermal conductivity $1.45 \text{ W/(m}\cdot\text{K)}$

Conductive thermal conductivity $1.41 \text{ W/(m}\cdot\text{K)}$

Radiative thermal conductivity $0.037 \text{ W/(m}\cdot\text{K)}$

Conductive thermal conductivity balance error 9.240271880983641E-10
Skeleton layer thickness 1.8e-04 m
Pore diameter 5.99e-05 m
Diffusion flame heat flux density 23,856.68 W/m²
Diffusion flame height 12,353.32 μm
Heat flux density to surface within skeletal structure 5.88e+06 W/m²
Heat flux density to surface outside skeletal structure 7,537.14 W/m²
Total heat flux density to surface 5.91e+06 W/m²
Binder sublimation heat flux density 5.91e+06 W/m²
Surface heat balance error 5.85e-04 W/m²

At pressure 4.06 MPa

Experimental burning rate 25.92 mm/s
Calculated burning rate 26.19 mm/s

First Conditional Fuel

Surface temperature 703.13 K
Linear burning rate 56.15 mm/s
Mass decomposition rate 93.15 kg·s⁻¹·m⁻²
Kinetic flame heat flux density to surface 1.18e+08 W/m²
Kinetic flame height 2.54 μm
Average kinetic flame density 7.3 kg/m³
Average kinetic flame temperature 2,205.07 K
Binder sublimation heat flux density 1.18e+08 W/m²
Surface heat balance error -0.019 W/m²

Secondary Conditional Fuel

Surface temperature 653.22 K
Linear burning rate 8.06 mm/s
Mass decomposition rate 13.38 kg·s⁻¹·m⁻²

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 16,669.69 W/m²
Kinetic flame height 2,332.26 μm
Average kinetic flame density 18.99 kg/m³
Average kinetic flame temperature 847.61 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 6.4e+06 W/m²
Kinetic flame height 9.18 μm
Average kinetic flame density 17 kg/m³
Average kinetic flame temperature 947.11 K
Heat flux density due to metal combustion, 1.3e+07 W/m²
Effective thermal conductivity 1.42 W/(m·K)
Conductive thermal conductivity 1.41 W/(m·K)
Radiative thermal conductivity 0.014 W/(m·K)
Conductive thermal conductivity balance error 9.240271880983641E-10
Skeleton layer thickness 7.11e-05 m
Pore diameter 2.37e-05 m
Diffusion flame heat flux density 60,320.37 W/m²
Diffusion flame height 4,852.06 μm
Heat flux density to surface within skeletal structure 1.02e+07 W/m²
Heat flux density to surface outside skeletal structure 7,918.44 W/m²
Total heat flux density to surface 1.02e+07 W/m²
Binder sublimation heat flux density 1.02e+07 W/m²
Surface heat balance error 0.001 W/m²

At pressure 6.5 MPa

Experimental burning rate 36.06 mm/s
Calculated burning rate 35.42 mm/s

First Conditional Fuel

Surface temperature 718.44 K
Linear burning rate 96.46 mm/s

Mass decomposition rate $160.03 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
Kinetic flame heat flux density to surface $2.07\text{e}+08 \text{ W/m}^2$
Kinetic flame height $1.44 \text{ }\mu\text{m}$
Average kinetic flame density 11.66 kg/m^3
Average kinetic flame temperature $2,212.72 \text{ K}$
Binder sublimation heat flux density $2.07\text{e}+08 \text{ W/m}^2$
Surface heat balance error -0.039 W/m^2

Secondary Conditional Fuel

Surface temperature 659.94 K
Linear burning rate 10.65 mm/s
Mass decomposition rate $17.67 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $17,178.36 \text{ W/m}^2$
Kinetic flame height $2,224.09 \text{ }\mu\text{m}$
Average kinetic flame density 30.32 kg/m^3
Average kinetic flame temperature 850.97 K

Skeleton Structure Region

Kinetic flame heat flux density to surface $6.74\text{e}+06 \text{ W/m}^2$
Kinetic flame height $8.62 \text{ }\mu\text{m}$
Average kinetic flame density 27.14 kg/m^3
Average kinetic flame temperature 950.47 K
Heat flux density due to metal combustion, $1.72\text{e}+07 \text{ W/m}^2$
Effective thermal conductivity $1.42 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity $1.41 \text{ W/(m}\cdot\text{K)}$
Radiative thermal conductivity $0.011 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity balance error $9.240271880983641\text{E}-10$
Skeleton layer thickness $5.3\text{e}-05 \text{ m}$
Pore diameter $1.77\text{e}-05 \text{ m}$

Diffusion flame heat flux density $46,309.04 \text{ W/m}^2$
Diffusion flame height $6,305.6 \text{ }\mu\text{m}$
Heat flux density to surface within skeletal structure $1.25\text{e}+07 \text{ W/m}^2$
Heat flux density to surface outside skeletal structure $8,215.34 \text{ W/m}^2$
Total heat flux density to surface $1.25\text{e}+07 \text{ W/m}^2$
Binder sublimation heat flux density $1.25\text{e}+07 \text{ W/m}^2$
Surface heat balance error $3.7\text{e}-04 \text{ W/m}^2$

Propellant Composition "Bas_1"

At pressure 1 MPa

Experimental burning rate 14.26 mm/s
Calculated burning rate 14.27 mm/s

First Conditional Fuel

Surface temperature 687.11 K
Linear burning rate 11.03 mm/s
Mass decomposition rate $18.3 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
Kinetic flame heat flux density to surface $2.28\text{e}+07 \text{ W/m}^2$
Kinetic flame height $13.22 \text{ }\mu\text{m}$
Average kinetic flame density 1.81 kg/m^3
Average kinetic flame temperature $2,197.05 \text{ K}$
Binder sublimation heat flux density $2.28\text{e}+07 \text{ W/m}^2$
Surface heat balance error 0.0063 W/m^2

Secondary Conditional Fuel

Surface temperature 677.54 K
Linear burning rate 7.53 mm/s
Mass decomposition rate $12.5 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $506,453.91 \text{ W/m}^2$
Kinetic flame height $333.19 \text{ }\mu\text{m}$
Average kinetic flame density 2.61 kg/m^3

Average kinetic flame temperature 1,521.27 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 8.43e+06 W/m²

Kinetic flame height 10.84 μm

Average kinetic flame density 3.5 kg/m³

Average kinetic flame temperature 1,134.27 K

Heat flux density due to metal combustion, 1.86e+06 W/m²

Effective thermal conductivity 0.4 W/(m·K)

Conductive thermal conductivity 0.4 W/(m·K)

Radiative thermal conductivity 2.83e-07 W/(m·K)

Conductive thermal conductivity balance error -4.092294297874943E-08

Skeleton layer thickness 1.35e-04 m

Pore diameter 1.76e-10 m

Diffusion flame heat flux density 55,770.53 W/m²

Diffusion flame height 5,358.49 μm

Heat flux density to surface within skeletal structure 3.89e+06 W/m²

Heat flux density to surface outside skeletal structure 314,986.83 W/m²

Total heat flux density to surface 1.4e+07 W/m²

Binder sublimation heat flux density 1.4e+07 W/m²

Surface heat balance error 9.73e-04 W/m²

At pressure 4.06 MPa

Experimental burning rate 38.01 mm/s

Calculated burning rate 38.04 mm/s

First Conditional Fuel

Surface temperature 730.56 K

Linear burning rate 55.07 mm/s

Mass decomposition rate 91.36 kg·s⁻¹·m⁻²

Kinetic flame heat flux density to surface 1.2e+08 W/m²

Kinetic flame height 2.48 μm

Average kinetic flame density 7.25 kg/m³

Average kinetic flame temperature 2,218.78 K

Binder sublimation heat flux density 1.2e+08 W/m²

Surface heat balance error 0.03 W/m²

Secondary Conditional Fuel

Surface temperature 699.29 K

Linear burning rate 17.67 mm/s

Mass decomposition rate 29.32 kg·s⁻¹·m⁻²

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 532,146.05 W/m²

Kinetic flame height 313.02 μm

Average kinetic flame density 10.51 kg/m³

Average kinetic flame temperature 1,532.15 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 3.23e+07 W/m²

Kinetic flame height 2.76 μm

Average kinetic flame density 14.06 kg/m³

Average kinetic flame temperature 1,145.15 K

Heat flux density due to metal combustion, 4.42e+06 W/m²

Effective thermal conductivity 0.4 W/(m·K)

Conductive thermal conductivity 0.4 W/(m·K)

Radiative thermal conductivity 1.15e-07 W/(m·K)

Conductive thermal conductivity balance error -4.092294297874943E-08

Skeleton layer thickness 5.48e-05 m

Pore diameter 7.15e-11 m

Diffusion flame heat flux density 144,530.82 W/m²

Diffusion flame height 2,052.65 μm

Heat flux density to surface within skeletal structure 7.36e+06 W/m²

Heat flux density to surface outside skeletal structure 425,559.2 W/m²

Total heat flux density to surface $2.28 \times 10^7 \text{ W/m}^2$
Binder sublimation heat flux density $2.28 \times 10^7 \text{ W/m}^2$
Surface heat balance error 0.0026 W/m^2

At pressure 6.5 MPa

Experimental burning rate 52.88 mm/s
Calculated burning rate 52.69 mm/s

First Conditional Fuel

Surface temperature 746.45 K
Linear burning rate 94.6 mm/s
Mass decomposition rate $156.94 \text{ kg} \cdot \text{s}^{-1} \cdot \text{m}^{-2}$
Kinetic flame heat flux density to surface $2.1 \times 10^8 \text{ W/m}^2$
Kinetic flame height $1.41 \text{ } \mu\text{m}$
Average kinetic flame density 11.59 kg/m^3
Average kinetic flame temperature 2,226.73 K
Binder sublimation heat flux density $2.1 \times 10^8 \text{ W/m}^2$
Surface heat balance error -0.059 W/m^2

Secondary Conditional Fuel

Surface temperature 707.27 K
Linear burning rate 23.84 mm/s
Mass decomposition rate $39.55 \text{ kg} \cdot \text{s}^{-1} \cdot \text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $534,575.59 \text{ W/m}^2$
Kinetic flame height $310.1 \text{ } \mu\text{m}$
Average kinetic flame density 16.79 kg/m^3
Average kinetic flame temperature 1,536.13 K

Skeleton Structure Region

Kinetic flame heat flux density to surface $5.02 \times 10^7 \text{ W/m}^2$
Kinetic flame height $1.76 \text{ } \mu\text{m}$
Average kinetic flame density 22.45 kg/m^3
Average kinetic flame temperature 1,149.13 K
Heat flux density due to metal combustion, $5.99 \times 10^6 \text{ W/m}^2$
Effective thermal conductivity $0.4 \text{ W/(m} \cdot \text{K)}$
Conductive thermal conductivity $0.4 \text{ W/(m} \cdot \text{K)}$
Radiative thermal conductivity $8.38 \times 10^{-8} \text{ W/(m} \cdot \text{K)}$
Conductive thermal conductivity balance error $-4.092294297874943 \times 10^{-8}$
Skeleton layer thickness $3.99 \times 10^{-5} \text{ m}$
Pore diameter $5.21 \times 10^{-11} \text{ m}$
Diffusion flame heat flux density $108,590.4 \text{ W/m}^2$
Diffusion flame height $2,724.67 \text{ } \mu\text{m}$
Heat flux density to surface within skeletal structure $9.18 \times 10^6 \text{ W/m}^2$
Heat flux density to surface outside skeletal structure $447,298.44 \text{ W/m}^2$
Total heat flux density to surface $2.83 \times 10^7 \text{ W/m}^2$
Binder sublimation heat flux density $2.83 \times 10^7 \text{ W/m}^2$
Surface heat balance error -0.0052 W/m^2

Propellant Composition "Bas_0"

At pressure 1 MPa

Experimental burning rate 5.97 mm/s
Calculated burning rate 5.97 mm/s

First Conditional Fuel

Surface temperature 828.82 K
Linear burning rate 10.01 mm/s
Mass decomposition rate $16.61 \text{ kg} \cdot \text{s}^{-1} \cdot \text{m}^{-2}$
Kinetic flame heat flux density to surface $2.43 \times 10^7 \text{ W/m}^2$
Kinetic flame height $11.86 \text{ } \mu\text{m}$
Average kinetic flame density 1.75 kg/m^3
Average kinetic flame temperature 2,267.91 K

Binder sublimation heat flux density $2.43\text{e}+07 \text{ W/m}^2$
Surface heat balance error -0.0061 W/m^2

Secondary Conditional Fuel

Surface temperature 790.83 K
Linear burning rate 2.72 mm/s
Mass decomposition rate $4.51 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $1.47\text{e}+06 \text{ W/m}^2$
Kinetic flame height $106.82 \text{ }\mu\text{m}$
Average kinetic flame density 2.52 kg/m^3
Average kinetic flame temperature $1,577.91 \text{ K}$

Skeleton Structure Region

Kinetic flame heat flux density to surface $342,515.73 \text{ W/m}^2$
Kinetic flame height $233.62 \text{ }\mu\text{m}$
Average kinetic flame density 3.33 kg/m^3
Average kinetic flame temperature $1,190.91 \text{ K}$
Heat flux density due to metal combustion, $1.52\text{e}+06 \text{ W/m}^2$
Effective thermal conductivity $0.4 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity $0.4 \text{ W/(m}\cdot\text{K)}$
Radiative thermal conductivity $8.37\text{e}-07 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity balance error $-4.092294297874943\text{E}-08$
Skeleton layer thickness $1.35\text{e}-04 \text{ m}$
Pore diameter $5.2\text{e}-10 \text{ m}$

Diffusion flame heat flux density $148,637.24 \text{ W/m}^2$
Diffusion flame height $1,934.36 \text{ }\mu\text{m}$
Heat flux density to surface within skeletal structure $482,205.12 \text{ W/m}^2$
Heat flux density to surface outside skeletal structure $1.09\text{e}+06 \text{ W/m}^2$
Total heat flux density to surface $5.64\text{e}+06 \text{ W/m}^2$
Binder sublimation heat flux density $5.64\text{e}+06 \text{ W/m}^2$
Surface heat balance error -0.0011 W/m^2

At pressure 4.06 MPa

Experimental burning rate 13.63 mm/s
Calculated burning rate 13.62 mm/s

First Conditional Fuel

Surface temperature 880.95 K
Linear burning rate 49.84 mm/s
Mass decomposition rate $82.69 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
Kinetic flame heat flux density to surface $1.27\text{e}+08 \text{ W/m}^2$
Kinetic flame height $2.22 \text{ }\mu\text{m}$
Average kinetic flame density 7.02 kg/m^3
Average kinetic flame temperature $2,293.98 \text{ K}$
Binder sublimation heat flux density $1.27\text{e}+08 \text{ W/m}^2$
Surface heat balance error -0.021 W/m^2

Secondary Conditional Fuel

Surface temperature 812.68 K
Linear burning rate 5.84 mm/s
Mass decomposition rate $9.69 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $1.69\text{e}+06 \text{ W/m}^2$
Kinetic flame height $92.06 \text{ }\mu\text{m}$
Average kinetic flame density 10.13 kg/m^3
Average kinetic flame temperature $1,588.84 \text{ K}$

Skeleton Structure Region

Kinetic flame heat flux density to surface $162,498.91 \text{ W/m}^2$
Kinetic flame height $478.97 \text{ }\mu\text{m}$
Average kinetic flame density 13.39 kg/m^3
Average kinetic flame temperature $1,201.84 \text{ K}$
Heat flux density due to metal combustion, $3.27\text{e}+06 \text{ W/m}^2$

Effective thermal conductivity 0.4 W/(m·K)
Conductive thermal conductivity 0.4 W/(m·K)
Radiative thermal conductivity 3.73e-07 W/(m·K)
Conductive thermal conductivity balance error -4.092294297874943E-08
Skeleton layer thickness 6.01e-05 m
Pore diameter 2.32e-10 m
Diffusion flame heat flux density 420,485.22 W/m²
Diffusion flame height 678.58 μm
Heat flux density to surface within skeletal structure 643,809.54 W/m²
Heat flux density to surface outside skeletal structure 1.37e+06 W/m²
Total heat flux density to surface 7.85e+06 W/m²
Binder sublimation heat flux density 7.85e+06 W/m²
Surface heat balance error -0.0017 W/m²

At pressure 6.5 MPa

Experimental burning rate 18 mm/s
Calculated burning rate 18.05 mm/s

First Conditional Fuel

Surface temperature 900 K
Linear burning rate 85.53 mm/s
Mass decomposition rate 141.9 kg·s⁻¹·m⁻²
Kinetic flame heat flux density to surface 2.22e+08 W/m²
Kinetic flame height 1.26 μm
Average kinetic flame density 11.2 kg/m³
Average kinetic flame temperature 2,303.5 K
Binder sublimation heat flux density 2.22e+08 W/m²
Surface heat balance error -0.029 W/m²

Secondary Conditional Fuel

Surface temperature 820.69 K
Linear burning rate 7.65 mm/s
Mass decomposition rate 12.7 kg·s⁻¹·m⁻²

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 1.74e+06 W/m²
Kinetic flame height 88.61 μm
Average kinetic flame density 16.2 kg/m³
Average kinetic flame temperature 1,592.85 K

Skeleton Structure Region

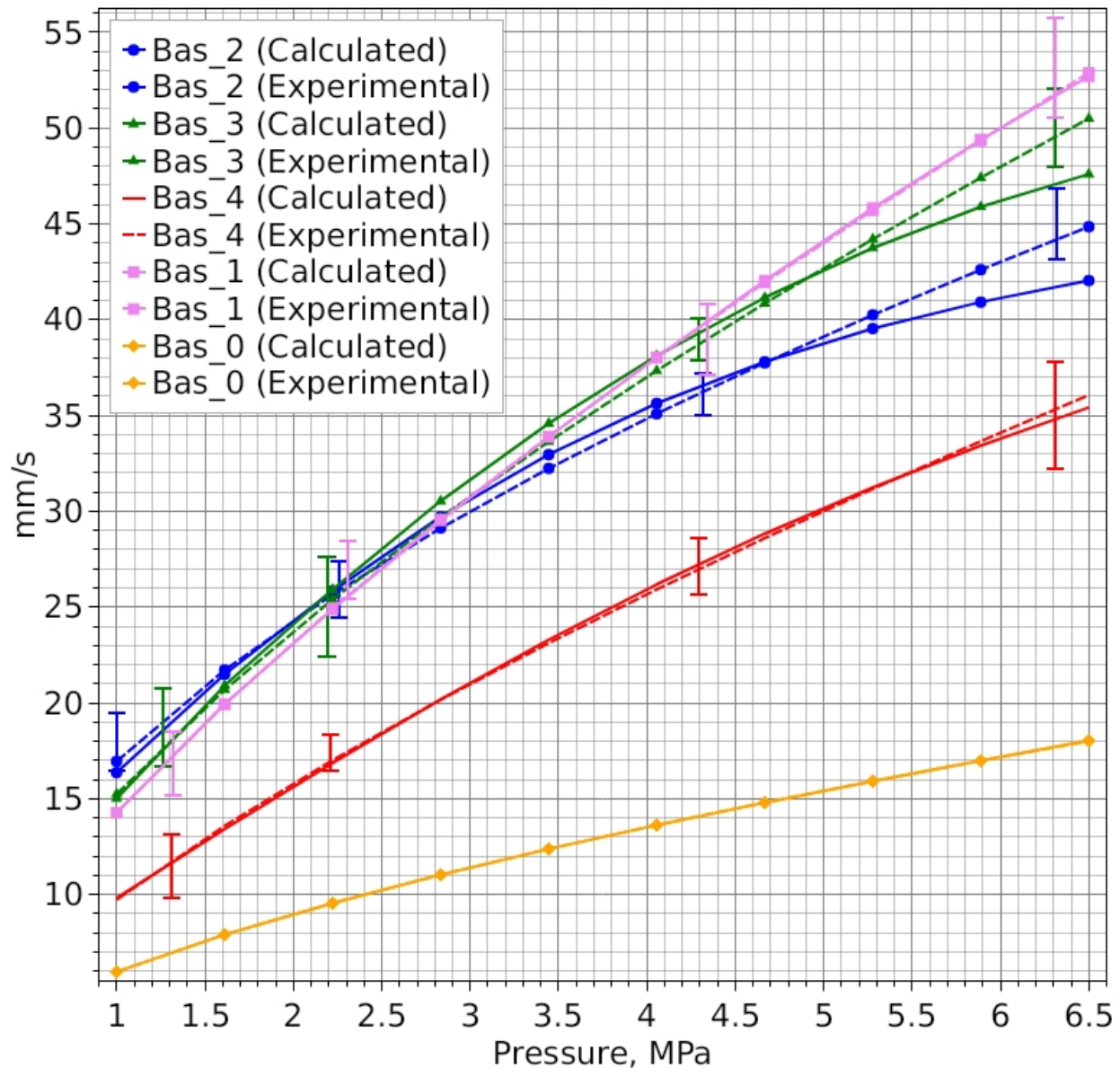
Kinetic flame heat flux density to surface 124,852 W/m²
Kinetic flame height 616.98 μm
Average kinetic flame density 21.39 kg/m³
Average kinetic flame temperature 1,205.85 K
Heat flux density due to metal combustion, 4.28e+06 W/m²
Effective thermal conductivity 0.4 W/(m·K)
Conductive thermal conductivity 0.4 W/(m·K)
Radiative thermal conductivity 2.8e-07 W/(m·K)
Conductive thermal conductivity balance error -4.092294297874943E-08
Skeleton layer thickness 4.52e-05 m
Pore diameter 1.74e-10 m
Diffusion flame heat flux density 325,333.09 W/m²
Diffusion flame height 874.58 μm
Heat flux density to surface within skeletal structure 555,305.69 W/m²
Heat flux density to surface outside skeletal structure 1.52e+06 W/m²
Total heat flux density to surface 9.26e+06 W/m²
Binder sublimation heat flux density 9.26e+06 W/m²
Surface heat balance error 0.0018 W/m²

1 MPa										
Parameter	Bas_2		Bas_3		Bas_4		Bas_1		Bas_0	
Experimental Burning Rate	16.94 mm/s		15.24 mm/s		9.73 mm/s		14.26 mm/s		5.97 mm/s	
Calculated Burning Rate	16.37 mm/s		15.01 mm/s		9.82 mm/s		14.27 mm/s		5.97 mm/s	
Difference	0.57 mm/s		0.23 mm/s		-0.09 mm/s		-0.01 mm/s		-7.05e-04 mm/s	
-	I	II	I	II	I	II	I	II	I	II
Linear Burning Rate	11.24 mm/s	8.97 mm/s	11.24 mm/s	11.24 mm/s	11.24 mm/s	3.35 mm/s	11.03 mm/s	7.53 mm/s	10.01 mm/s	2.72 mm/s
Surface Temperature	661.26 K	655.78 K	661.26 K	661.26 K	661.26 K	632.91 K	687.11 K	677.54 K	828.82 K	790.83 K
Kinetic Flame Heat Flux Density (Inter-Pocket)	2.26e+07 W/m ²		2.26e+07 W/m ²		2.26e+07 W/m ²		2.28e+07 W/m ²		2.43e+07 W/m ²	
Kinetic Flame Heat Flux Density (Skeleton)	8.48e+06 W/m ²		5.94e+07 W/m ²		5.75e+06 W/m ²		8.43e+06 W/m ²		342,515.73 W/m ²	
Kinetic Flame Height (Skeleton)	11.03 μm		3.3 μm		10.57 μm		10.84 μm		233.62 μm	
Kinetic Flame Heat Flux Density (Out-Skeleton)	420,755.22 W/m ²		336,619.14 W/m ²		15,994.54 W/m ²		506,453.91 W/m ²		1.47e+06 W/m ²	
Kinetic Flame Height (Out-Skeleton)	406.23 μm		506.13 μm		2,557.69 μm		333.19 μm		106.82 μm	
Diffusion Flame Heat Flux Density	46,219.76 W/m ²		94,261.87 W/m ²		23,856.68 W/m ²		55,770.53 W/m ²		148,637.24 W/m ²	
Diffusion Flame Height	6,378.71 μm		3,196.14 μm		12,353.32 μm		5,358.49 μm		1,934.36 μm	
Metal Combustion Heat Flux Density	4.44e+06 W/m ²		4.94e+06 W/m ²		5.37e+06 W/m ²		1.86e+06 W/m ²		1.52e+06 W/m ²	

4.06 MPa										
Parameter	Bas_2		Bas_3		Bas_4		Bas_1		Bas_0	
Experimental Burning Rate	35.09 mm/s		37.34 mm/s		25.92 mm/s		38.01 mm/s		13.63 mm/s	
Calculated Burning Rate	35.63 mm/s		38.11 mm/s		26.19 mm/s		38.04 mm/s		13.62 mm/s	
Difference	-0.54 mm/s		-0.77 mm/s		-0.26 mm/s		-0.033 mm/s		0.0059 mm/s	
-	I	II	I	II	I	II	I	II	I	II
Linear Burning Rate	56.15 mm/s	16.37 mm/s	56.15 mm/s	20.83 mm/s	56.15 mm/s	8.06 mm/s	55.07 mm/s	17.67 mm/s	49.84 mm/s	5.84 mm/s
Surface Temperature	703.13 K	670.59 K	703.13 K	676.71 K	703.13 K	653.22 K	730.56 K	699.29 K	880.95 K	812.68 K
Kinetic Flame Heat Flux Density (Inter-Pocket)	1.18e+08 W/m ²		1.18e+08 W/m ²		1.18e+08 W/m ²		1.2e+08 W/m ²		1.27e+08 W/m ²	
Kinetic Flame Heat Flux Density (Skeleton)	1.23e+07 W/m ²		8.39e+07 W/m ²		6.4e+06 W/m ²		3.23e+07 W/m ²		162,498.91 W/m ²	
Kinetic Flame Height (Skeleton)	7.5 μm		2.32 μm		9.18 μm		2.76 μm		478.97 μm	
Kinetic Flame Heat Flux Density (Out-Skeleton)	566,468.77 W/m ²		446,592.54 W/m ²		16,669.69 W/m ²		532,146.05 W/m ²		1.69e+06 W/m ²	
Kinetic Flame Height (Out-Skeleton)	299.12 μm		378.04 μm		2,332.26 μm		313.02 μm		92.06 μm	
Diffusion Flame Heat Flux Density	154,276.59 W/m ²		309,950.8 W/m ²		60,320.37 W/m ²		144,530.82 W/m ²		420,485.22 W/m ²	
Diffusion Flame Height	1,901.4 μm		967.02 μm		4,852.06 μm		2,052.65 μm		678.58 μm	
Metal Combustion Heat Flux Density	7.9e+06 W/m ²		7.36e+06 W/m ²		1.3e+07 W/m ²		4.42e+06 W/m ²		3.27e+06 W/m ²	

6.5 MPa										
Parameter	Bas_2		Bas_3		Bas_4		Bas_1		Bas_0	
Experimental Burning Rate	44.85 mm/s		50.5 mm/s		36.06 mm/s		52.88 mm/s		18 mm/s	
Calculated Burning Rate	42.03 mm/s		47.6 mm/s		35.42 mm/s		52.69 mm/s		18.05 mm/s	
Difference	2.81 mm/s		2.9 mm/s		0.64 mm/s		0.19 mm/s		-0.048 mm/s	
-	I	II	I	II	I	II	I	II	I	II
Linear Burning Rate	96.46 mm/s	18.58 mm/s	96.46 mm/s	24.16 mm/s	96.46 mm/s	10.65 mm/s	94.6 mm/s	23.84 mm/s	85.53 mm/s	7.65 mm/s
Surface Temperature	718.44 K	673.8 K	718.44 K	680.54 K	718.44 K	659.94 K	746.45 K	707.27 K	900 K	820.69 K
Kinetic Flame Heat Flux Density (Inter-Pocket)	2.07e+08 W/m ²		2.07e+08 W/m ²		2.07e+08 W/m ²		2.1e+08 W/m ²		2.22e+08 W/m ²	
Kinetic Flame Heat Flux Density (Skeleton)	1.5e+07 W/m ²		1e+08 W/m ²		6.74e+06 W/m ²		5.02e+07 W/m ²		124,852 W/m ²	
Kinetic Flame Height (Skeleton)	6.13 μm		1.94 μm		8.62 μm		1.76 μm		616.98 μm	
Kinetic Flame Heat Flux Density (Out-Skeleton)	674,882.56 W/m ²		520,765.07 W/m ²		17,178.36 W/m ²		534,575.59 W/m ²		1.74e+06 W/m ²	
Kinetic Flame Height (Out-Skeleton)	250.59 μm		323.46 μm		2,224.09 μm		310.1 μm		88.61 μm	
Diffusion Flame Heat Flux Density	137,981.09 W/m ²		271,197.91 W/m ²		46,309.04 W/m ²		108,590.4 W/m ²		325,333.09 W/m ²	
Diffusion Flame Height	2,123.63 μm		1,103.79 μm		6,305.6 μm		2,724.67 μm		874.58 μm	
Metal Combustion Heat Flux Density	8.94e+06 W/m ²		8.22e+06 W/m ²		1.72e+07 W/m ²		5.99e+06 W/m ²		4.28e+06 W/m ²	

Burning Rates - Group Optimization (All Propellants)



Propellant Composition Parameters

Composition "Bas_2"

Density 1,659 kg/m³, specific heat capacity 1,500 J/kg.K, initial temperature 298 K, mass fraction of "pockets" in "binder-metal" composition 0.7.

Vieille's Power Law: $U = A \cdot p^v$, $A = 1.285341\text{E-}05$; $v = 0.52$.

Gas Phase Parameters in Inter-Pocket Medium

Thermal conductivity 0.1 W/(m·K)
 Average molar mass 0.033 kg/mol
 Specific heat capacity (constant volume) 635.6 J/kg.K
 Kinetic flame temperature 3,707 K

Gas Phase Parameters in Pocket Region

Thermal conductivity 0.1 W/(m·K)
 Average molar mass 0.033 kg/mol
 Specific heat capacity (constant volume) 635.6 J/kg.K
 Diffusion flame temperature 3,604 K

Gas Phase Parameters in Skeleton Structure

Thermal conductivity 0.1 W/(m·K)
 Average molar mass 0.033 kg/mol
 Specific heat capacity (constant volume) 635.6 J/kg.K
 Kinetic flame temperature 1,591 K

Gas Phase Parameters Outside Skeleton Structure

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 2,365 K

Composition "Bas_3"

Density 1,659 kg/m³, specific heat capacity 1,500 J/kg.K, initial temperature 298 K, mass fraction of "pockets" in "binder-metal" composition 0.57.

Vieille's Power Law: $U = A \cdot p^v$, $A = 2.203002E-06$; $v = 0.64$.

Gas Phase Parameters in Inter-Pocket Medium

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 3,707 K

Gas Phase Parameters in Pocket Region

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Diffusion flame temperature 3,674 K

Gas Phase Parameters in Skeleton Structure

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 2,623 K

Gas Phase Parameters Outside Skeleton Structure

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 2,365 K

Composition "Bas_4"

Density 1,659 kg/m³, specific heat capacity 1,500 J/kg.K, initial temperature 298 K, mass fraction of "pockets" in "binder-metal" composition 0.56.

Vieille's Power Law: $U = A \cdot p^v$, $A = 6.137894E-07$; $v = 0.7$.

Gas Phase Parameters in Inter-Pocket Medium

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 3,707 K

Gas Phase Parameters in Pocket Region

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Diffusion flame temperature 3,580 K

Gas Phase Parameters in Skeleton Structure

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 1,241 K

Gas Phase Parameters Outside Skeleton Structure

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 1,042 K

Composition "Bas_1"

Density 1,659 kg/m³, specific heat capacity 1,500 J/kg.K, initial temperature 298 K, mass fraction of "pockets" in "binder-metal" composition 0.7.

Vieille's Power Law: $U = A \cdot p^v$, $A = 9.000448E-07$; $v = 0.7$.

Gas Phase Parameters in Inter-Pocket Medium

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol

Specific heat capacity (constant volume) 635.6 J/kg.K

Kinetic flame temperature 3,707 K

Gas Phase Parameters in Pocket Region

Thermal conductivity 0.1 W/(m·K)

Average molar mass 0.033 kg/mol

Specific heat capacity (constant volume) 635.6 J/kg.K

Diffusion flame temperature 3,666 K

Gas Phase Parameters in Skeleton Structure

Thermal conductivity 0.1 W/(m·K)

Average molar mass 0.033 kg/mol

Specific heat capacity (constant volume) 635.6 J/kg.K

Kinetic flame temperature 1,591 K

Gas Phase Parameters Outside Skeleton Structure

Thermal conductivity 0.1 W/(m·K)

Average molar mass 0.033 kg/mol

Specific heat capacity (constant volume) 635.6 J/kg.K

Kinetic flame temperature 2,365 K

Composition "Bas_0"

Density 1,659 kg/m³, specific heat capacity 1,500 J/kg.K, initial temperature 298 K, mass fraction of "pockets" in "binder-metal" composition 0.7.

Vieille's Power Law: $U = A \cdot p^v$, $A = 1.720582E-06$; $v = 0.59$.

Gas Phase Parameters in Inter-Pocket Medium

Thermal conductivity 0.1 W/(m·K)

Average molar mass 0.033 kg/mol

Specific heat capacity (constant volume) 635.6 J/kg.K

Kinetic flame temperature 3,707 K

Gas Phase Parameters in Pocket Region

Thermal conductivity 0.1 W/(m·K)

Average molar mass 0.033 kg/mol

Specific heat capacity (constant volume) 635.6 J/kg.K

Diffusion flame temperature 3,666 K

Gas Phase Parameters in Skeleton Structure

Thermal conductivity 0.1 W/(m·K)

Average molar mass 0.033 kg/mol

Specific heat capacity (constant volume) 635.6 J/kg.K

Kinetic flame temperature 1,591 K

Gas Phase Parameters Outside Skeleton Structure

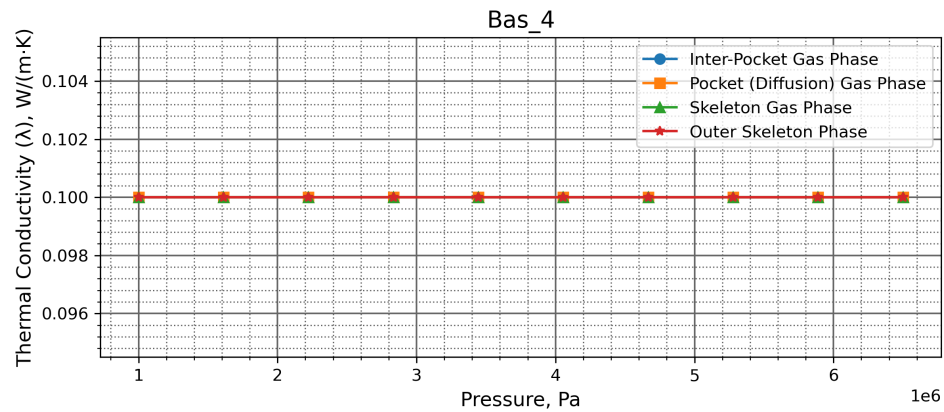
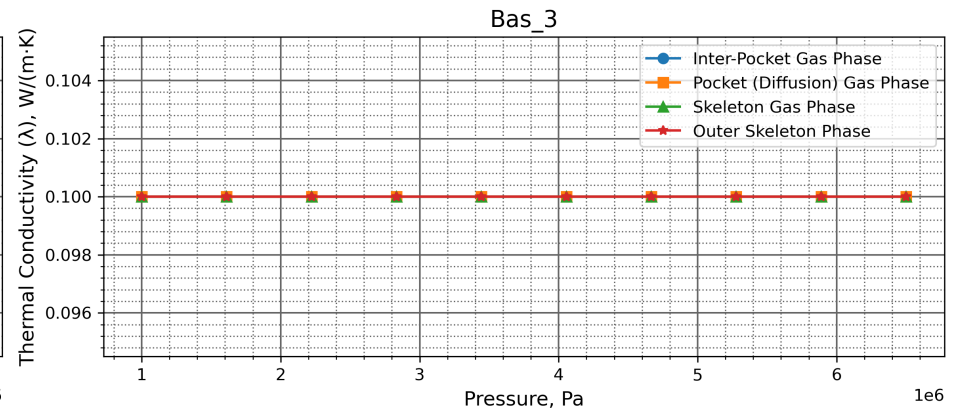
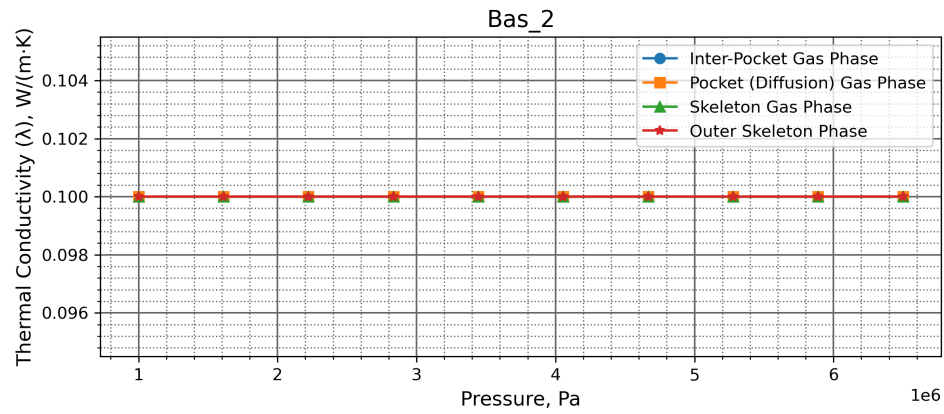
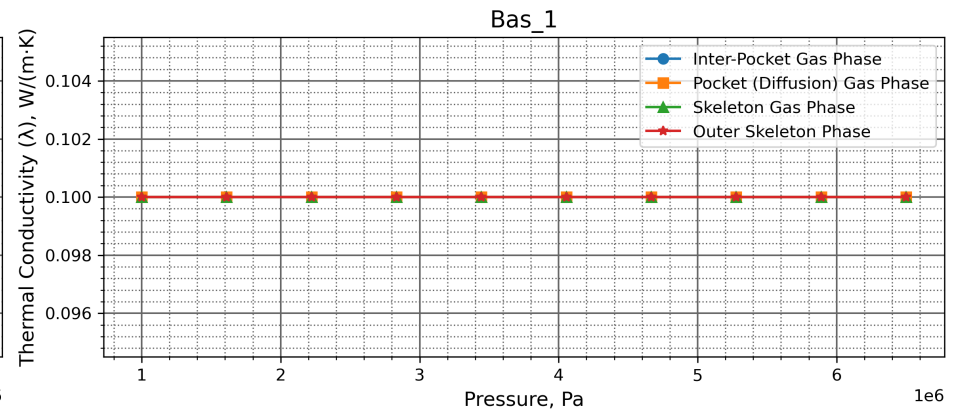
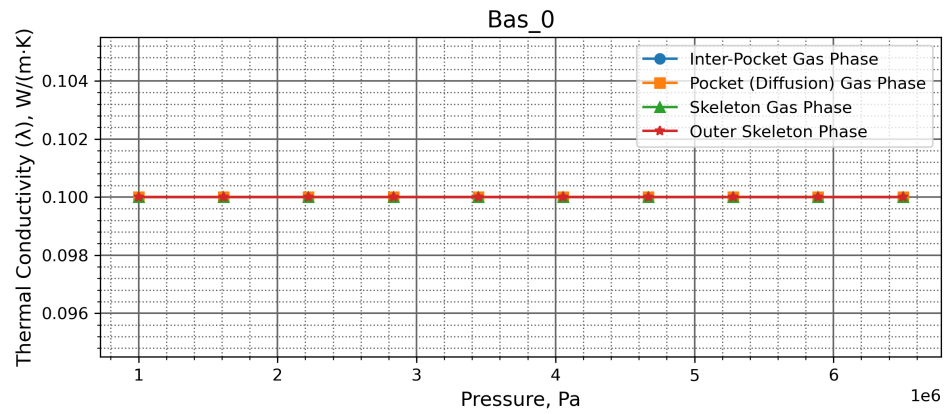
Thermal conductivity 0.1 W/(m·K)

Average molar mass 0.033 kg/mol

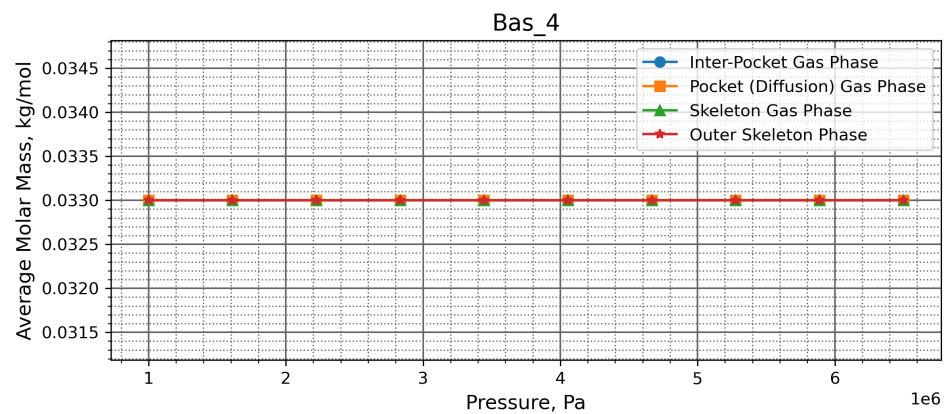
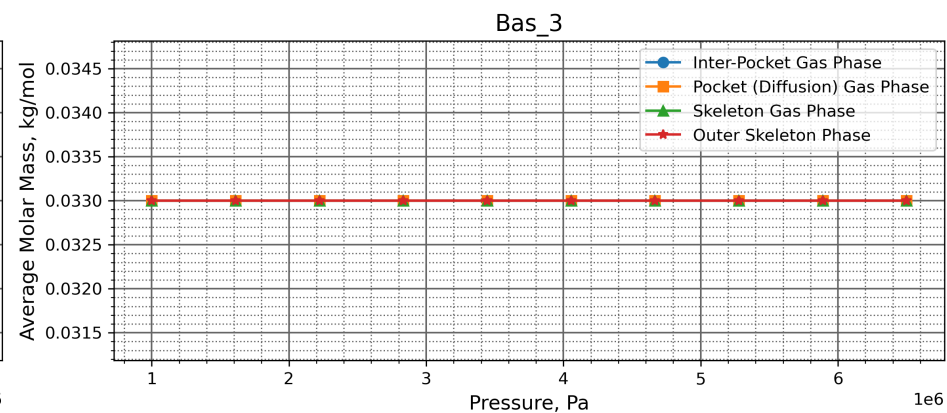
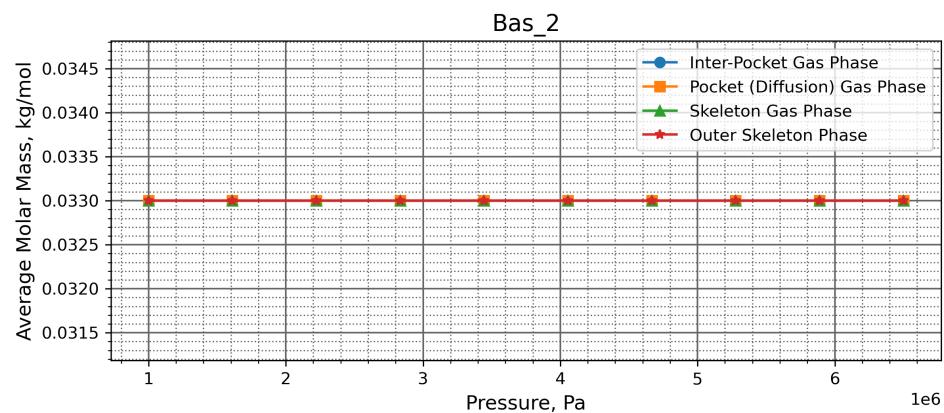
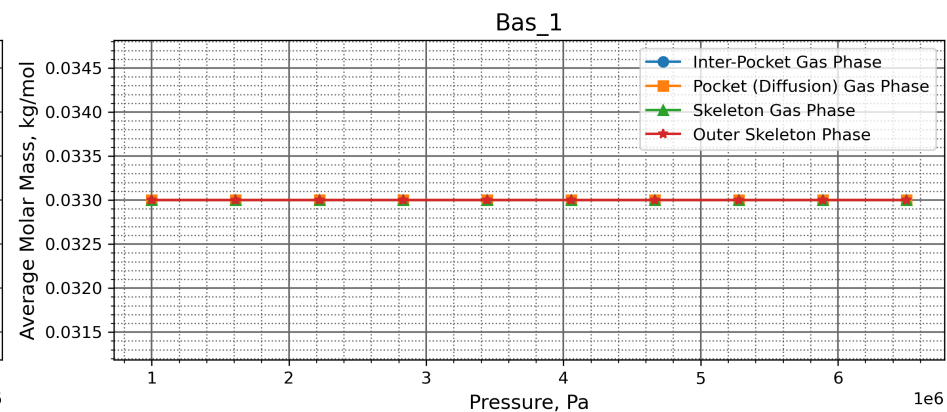
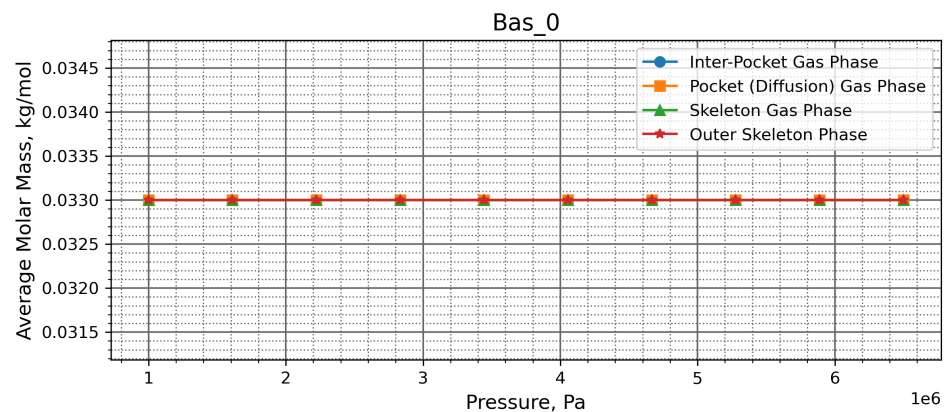
Specific heat capacity (constant volume) 635.6 J/kg.K

Kinetic flame temperature 2,365 K

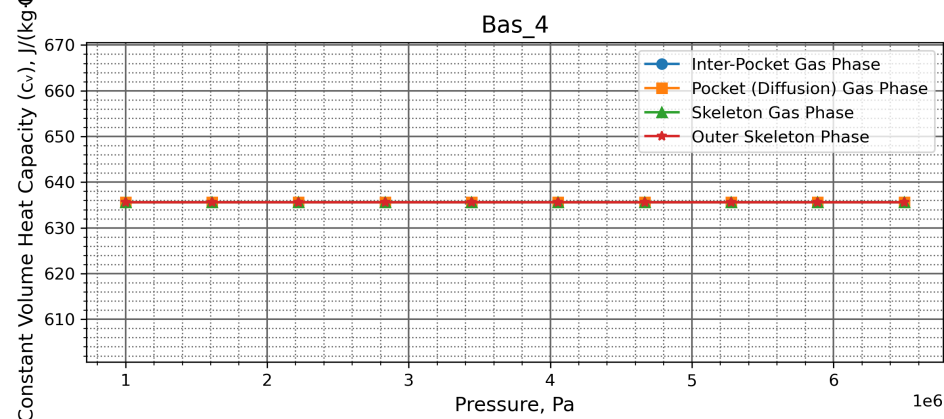
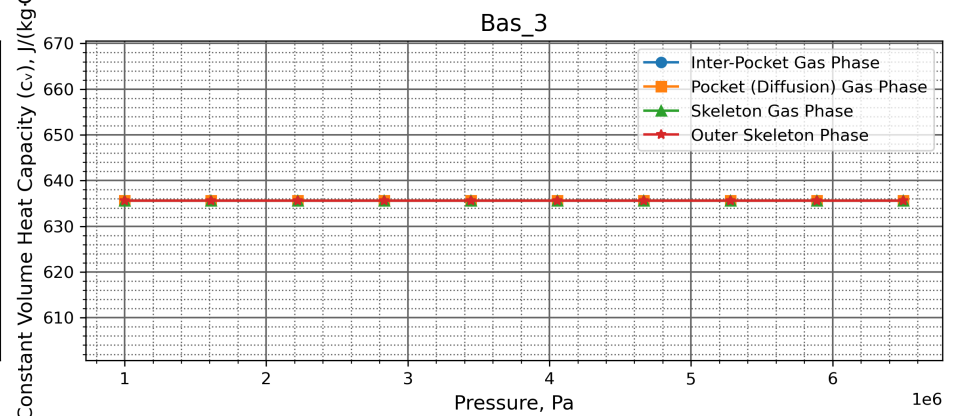
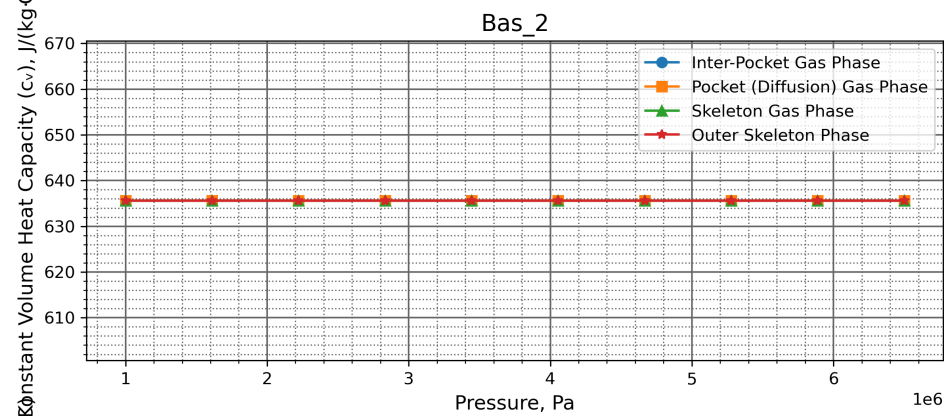
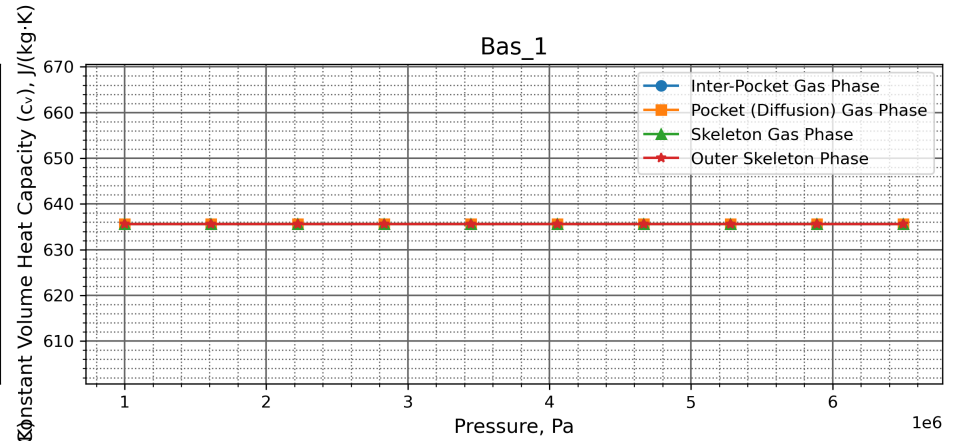
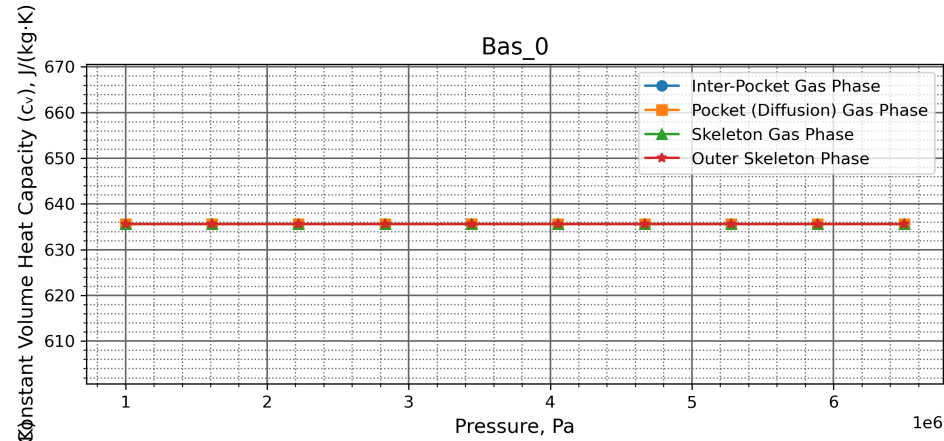
Thermal Conductivity (λ), W/(m·K)



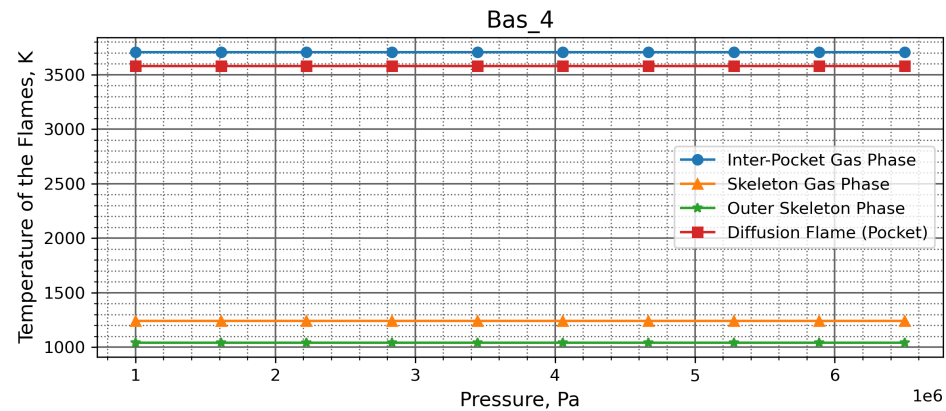
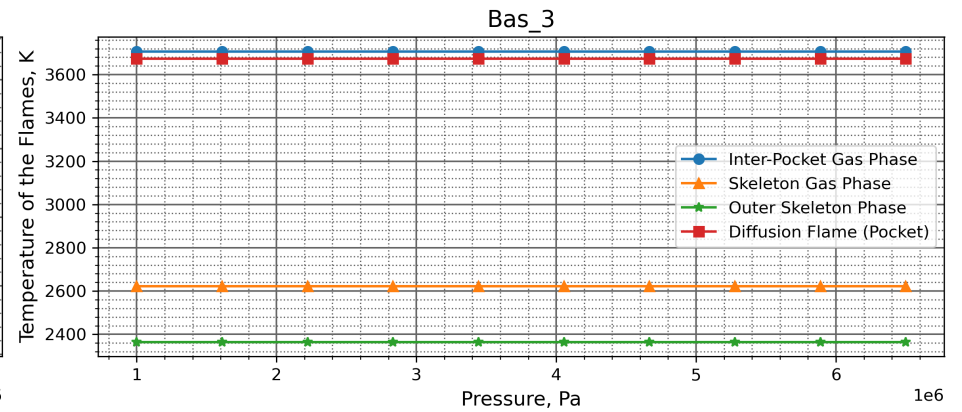
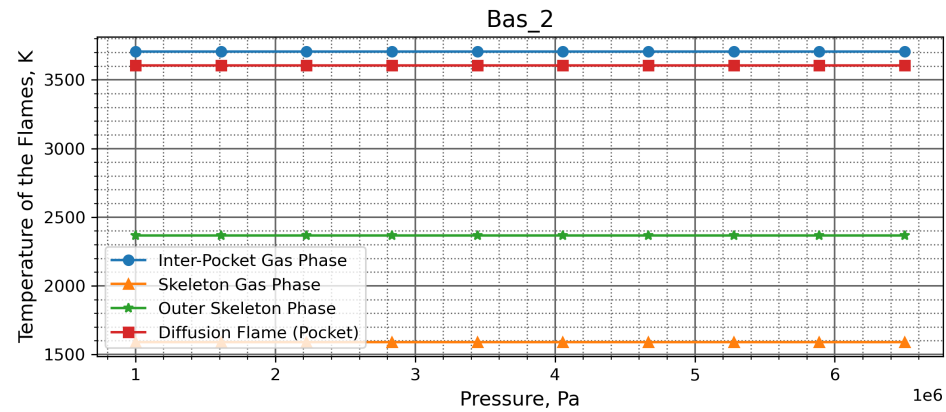
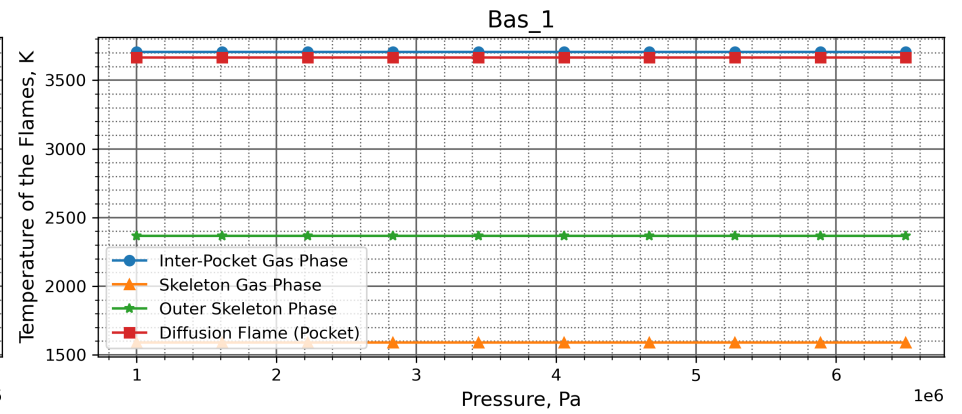
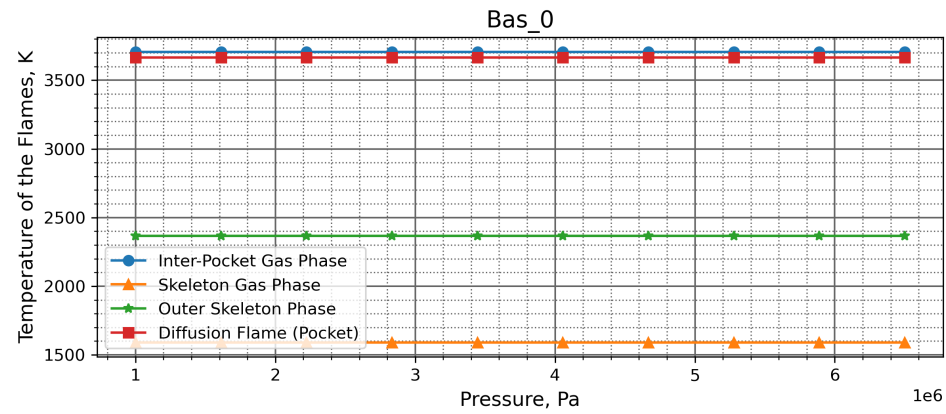
Average Molar Mass, kg/mol



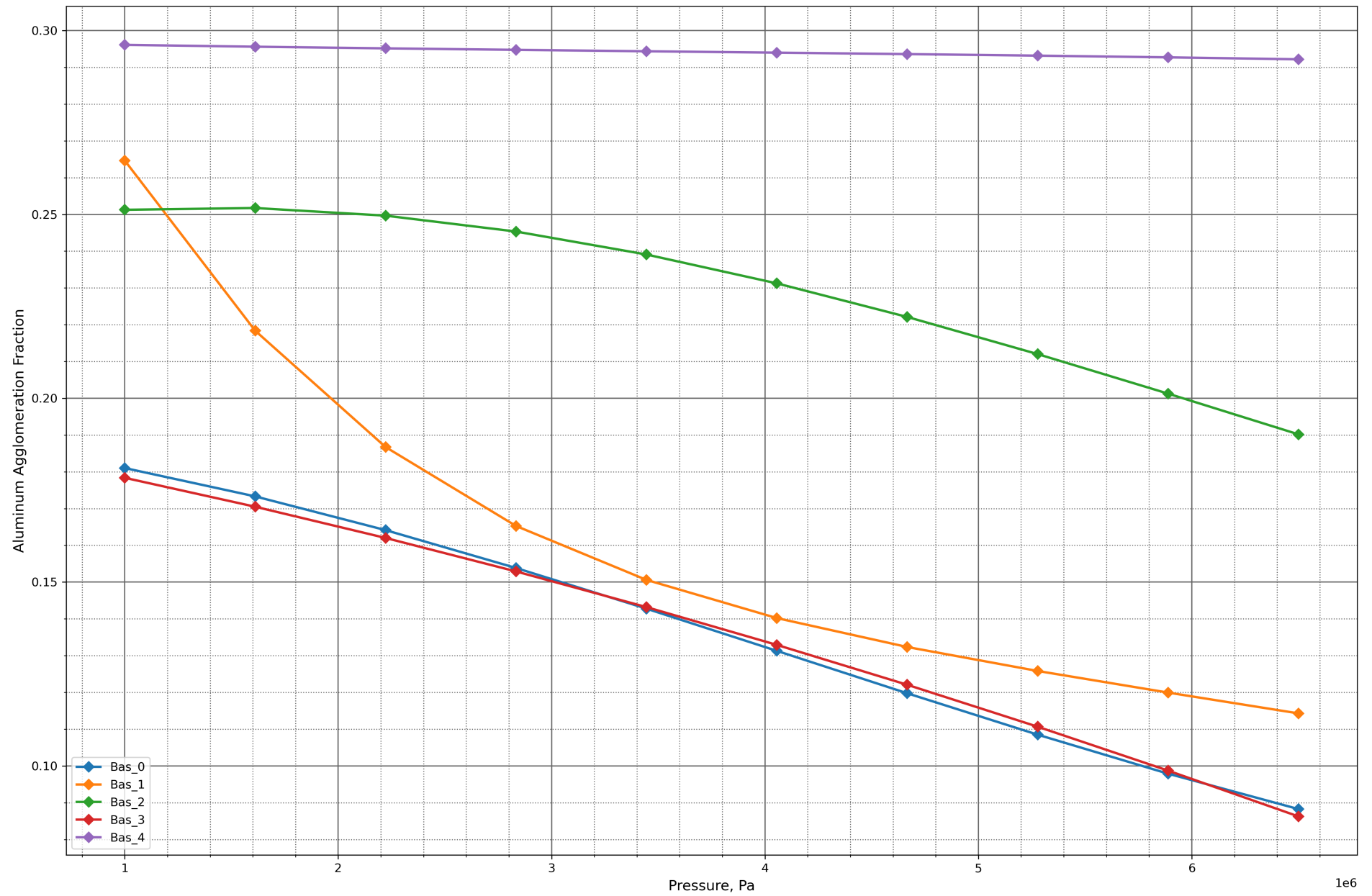
Constant Volume Heat Capacity (c_v), J/(kg·K)



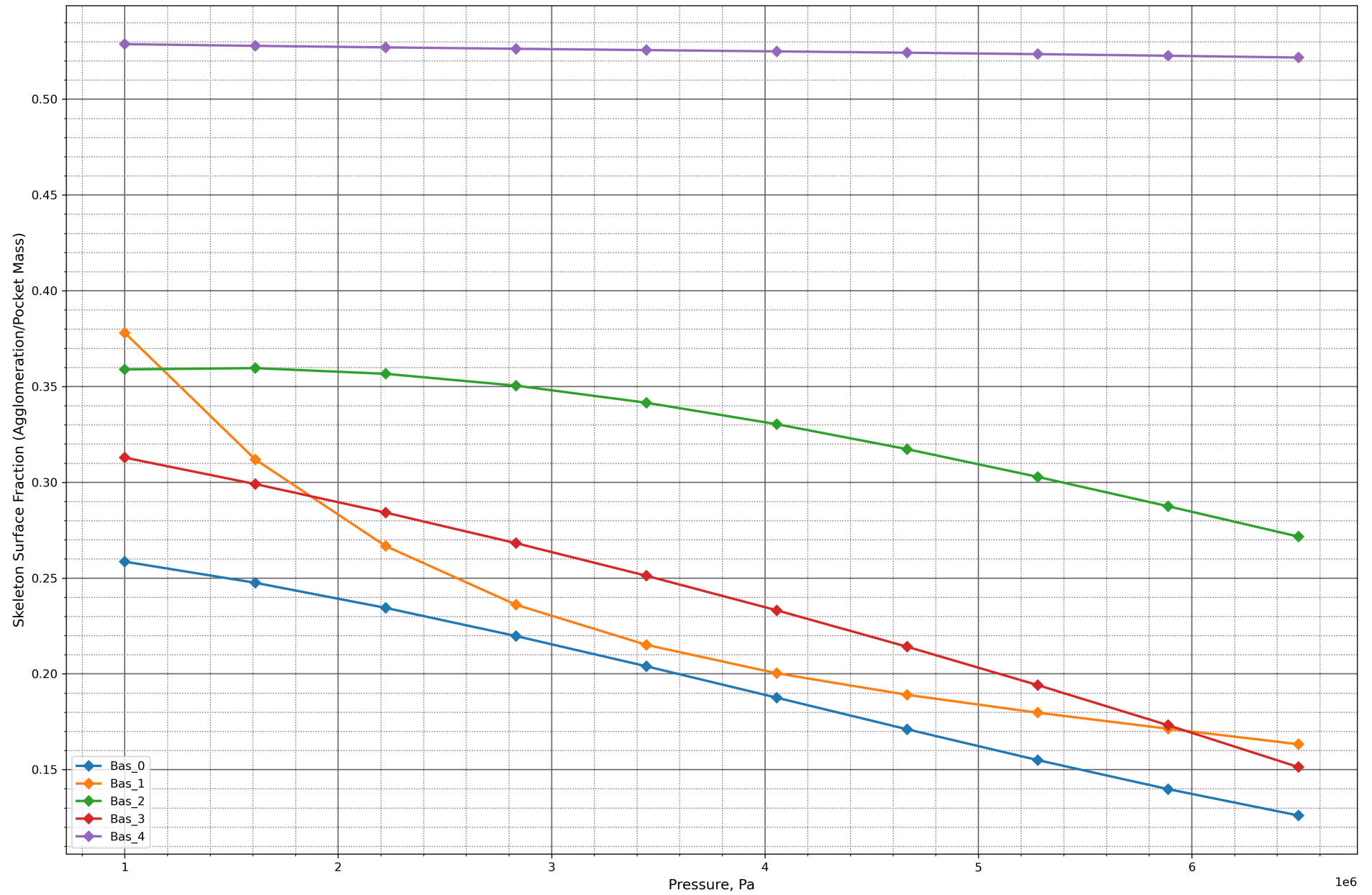
Temperature of the Flames, K



Aluminum Agglomeration Fraction



Skeleton Surface Fraction (Agglomeration/Pocket Mass)



Performance Report

System Information

Operating System: Unix 6.8.0.124 (Ubuntu 24.04.4 LTS)
Processor: Intel(R) Xeon(R) CPU E5-2697 v4 @ 2.30GHz (Base Clock Frequency 2.29 GHz)
Memory: 15.6 GB (64-bit), Unknown
Machine Name: vscode
Runtime: .NET 10.0.9 (.NET 10.0.9)
Architecture: X64 / X64

Total Execution Time

Total execution time: 09:01:47.728

Garbage Collection Information

Total Memory: 7672320 bytes
GC Pause Time Percentage: 0.17%
Generation 0 Collections: 41413
Generation 1 Collections: 3229
Generation 2 Collections: 323

Detailed Performance Metrics

Method: Telemetry/Proc/0/OptimizationProblemExecutionPerformance

Call count: 15957583
Maximum execution time: 25.1491 ms
Minimum execution time: 0.0001 ms
Mean execution time: 1.0956942258485942 ms
Execution time standard deviation: 0.25921901565210637 ms

Method: Telemetry/Proc/0/OptimizationProblemExecutionPerformance

Call count: 15016607
Maximum execution time: 21.6833 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.40618261282332624 ms
Execution time standard deviation: 0.0715761522545499 ms

Method: Telemetry/Proc/0/OptimizationProblemExecutionPerformance

Call count: 14684298
Maximum execution time: 25.7367 ms
Minimum execution time: 0.0001 ms
Mean execution time: 0.3755554662129553 ms
Execution time standard deviation: 0.12070912280752234 ms

Method: Telemetry/Proc/1/OptimizationProblemExecutionPerformance

Call count: 15857070
Maximum execution time: 19.1387 ms
Minimum execution time: 0.0002 ms
Mean execution time: 1.0927440869716487 ms
Execution time standard deviation: 0.2547477456695454 ms

Method: Telemetry/Proc/1/OptimizationProblemExecutionPerformance

Call count: 14862340
Maximum execution time: 13.7547 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.40627502782872 ms
Execution time standard deviation: 0.06668512430796947 ms

Method: Telemetry/Proc/1/OptimizationProblemExecutionPerformance

Call count: 14534714
Maximum execution time: 16.172 ms

Minimum execution time: 0.0002 ms
Mean execution time: 0.376474360878362 ms
Execution time standard deviation: 0.11788911153011032 ms

Method: Telemetry/Proc/2/OptimizationProblemExecutionPerformance
Call count: 15857070
Maximum execution time: 28.4727 ms
Minimum execution time: 0.0002 ms
Mean execution time: 1.0806378723622114 ms
Execution time standard deviation: 0.2636628076867694 ms

Method: Telemetry/Proc/2/OptimizationProblemExecutionPerformance
Call count: 14802714
Maximum execution time: 25.4191 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.4046304619883351 ms
Execution time standard deviation: 0.0704986710248423 ms

Method: Telemetry/Proc/2/OptimizationProblemExecutionPerformance
Call count: 14457500
Maximum execution time: 18.1728 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.37391461767939593 ms
Execution time standard deviation: 0.12017133236243395 ms

Method: Telemetry/Proc/3/OptimizationProblemExecutionPerformance
Call count: 15857070
Maximum execution time: 29.1887 ms
Minimum execution time: 0.0002 ms
Mean execution time: 1.0968812519776097 ms
Execution time standard deviation: 0.2544954876736725 ms

Method: Telemetry/Proc/3/OptimizationProblemExecutionPerformance
Call count: 14854081
Maximum execution time: 16.074 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.40823175119341665 ms
Execution time standard deviation: 0.06466430033242454 ms

Method: Telemetry/Proc/3/OptimizationProblemExecutionPerformance
Call count: 14519620
Maximum execution time: 18.4361 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.37575215715014815 ms
Execution time standard deviation: 0.1185932173352746 ms

Method: Telemetry/Proc/4/OptimizationProblemExecutionPerformance
Call count: 15857070
Maximum execution time: 26.2197 ms
Minimum execution time: 0.0002 ms
Mean execution time: 1.085208017054909 ms
Execution time standard deviation: 0.26549074550687646 ms

Method: Telemetry/Proc/4/OptimizationProblemExecutionPerformance
Call count: 14790159
Maximum execution time: 29.4465 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.4057529218786289 ms
Execution time standard deviation: 0.07154093186520823 ms

Method: Telemetry/Proc/4/OptimizationProblemExecutionPerformance

Call count: 14440609

Maximum execution time: 24.3141 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.3743878412814723 ms

Execution time standard deviation: 0.12111635073067321 ms

Method: Telemetry/Proc/5/OptimizationProblemExecutionPerformance

Call count: 15857070

Maximum execution time: 25.8037 ms

Minimum execution time: 0.0002 ms

Mean execution time: 1.0919873816033716 ms

Execution time standard deviation: 0.25677819136520247 ms

Method: Telemetry/Proc/5/OptimizationProblemExecutionPerformance

Call count: 14835319

Maximum execution time: 19.5868 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.4063609832657199 ms

Execution time standard deviation: 0.06729319233656399 ms

Method: Telemetry/Proc/5/OptimizationProblemExecutionPerformance

Call count: 14494615

Maximum execution time: 17.2711 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.37635245675033996 ms

Execution time standard deviation: 0.11758410205596727 ms

Method: Telemetry/Proc/6/OptimizationProblemExecutionPerformance

Call count: 15857070

Maximum execution time: 32.1687 ms

Minimum execution time: 0.0002 ms

Mean execution time: 1.0966323167518497 ms

Execution time standard deviation: 0.2631509289095171 ms

Method: Telemetry/Proc/6/OptimizationProblemExecutionPerformance

Call count: 14825657

Maximum execution time: 24.7706 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.40856922184292604 ms

Execution time standard deviation: 0.07023844360301848 ms

Method: Telemetry/Proc/6/OptimizationProblemExecutionPerformance

Call count: 14475844

Maximum execution time: 15.573 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.37718207935918446 ms

Execution time standard deviation: 0.12085959024792906 ms

Method: Telemetry/Proc/7/OptimizationProblemExecutionPerformance

Call count: 15857070

Maximum execution time: 24.7301 ms

Minimum execution time: 0.0002 ms

Mean execution time: 1.0946910947041386 ms

Execution time standard deviation: 0.2561404786642138 ms

Method: Telemetry/Proc/7/OptimizationProblemExecutionPerformance

Call count: 14836883

Maximum execution time: 16.9402 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.4084256637867304 ms
Execution time standard deviation: 0.06544681087794625 ms

Method: Telemetry/Proc/7/OptimizationProblemExecutionPerformance

Call count: 14509390
Maximum execution time: 18.5068 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.3766548547802512 ms
Execution time standard deviation: 0.11956187266518721 ms

Method: Telemetry/Proc/8/OptimizationProblemExecutionPerformance

Call count: 15857070
Maximum execution time: 28.5909 ms
Minimum execution time: 0.0002 ms
Mean execution time: 1.1055755100595186 ms
Execution time standard deviation: 0.24948751575182304 ms

Method: Telemetry/Proc/8/OptimizationProblemExecutionPerformance

Call count: 14914295
Maximum execution time: 28.7996 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.40945528791004915 ms
Execution time standard deviation: 0.06569039073997236 ms

Method: Telemetry/Proc/8/OptimizationProblemExecutionPerformance

Call count: 14578134
Maximum execution time: 37.8138 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.3796899341095358 ms
Execution time standard deviation: 0.11765637914815885 ms

Method: Telemetry/Proc/9/OptimizationProblemExecutionPerformance

Call count: 15857070
Maximum execution time: 25.9725 ms
Minimum execution time: 0.0002 ms
Mean execution time: 1.0895173142896408 ms
Execution time standard deviation: 0.2582324765657147 ms

Method: Telemetry/Proc/9/OptimizationProblemExecutionPerformance

Call count: 14828470
Maximum execution time: 16.1961 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.40549077629043806 ms
Execution time standard deviation: 0.06782355160842715 ms

Method: Telemetry/Proc/9/OptimizationProblemExecutionPerformance

Call count: 14491750
Maximum execution time: 20.4909 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.3743161450135287 ms
Execution time standard deviation: 0.11978216187051471 ms

Method: Telemetry/Proc/10/OptimizationProblemExecutionPerformance

Call count: 15857070
Maximum execution time: 28.3598 ms
Minimum execution time: 0.0002 ms
Mean execution time: 1.0985188094144547 ms

Execution time standard deviation: 0.26114270851693233 ms

Method: Telemetry/Proc/10/OptimizationProblemExecutionPerformance

Call count: 14810620

Maximum execution time: 23.8472 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.40903786323592795 ms

Execution time standard deviation: 0.06815963557695465 ms

Method: Telemetry/Proc/10/OptimizationProblemExecutionPerformance

Call count: 14459418

Maximum execution time: 20.4816 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.37729241327691426 ms

Execution time standard deviation: 0.11941220553401949 ms

Method: Telemetry/Proc/11/OptimizationProblemExecutionPerformance

Call count: 15857070

Maximum execution time: 22.6492 ms

Minimum execution time: 0.0002 ms

Mean execution time: 1.0903046582941733 ms

Execution time standard deviation: 0.25669620093104745 ms

Method: Telemetry/Proc/11/OptimizationProblemExecutionPerformance

Call count: 14833284

Maximum execution time: 28.3354 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.4059849759365198 ms

Execution time standard deviation: 0.06784914634920201 ms

Method: Telemetry/Proc/11/OptimizationProblemExecutionPerformance

Call count: 14485949

Maximum execution time: 21.8667 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.3767950224593993 ms

Execution time standard deviation: 0.12070752651970541 ms

Method: Telemetry/Proc/12/OptimizationProblemExecutionPerformance

Call count: 15857070

Maximum execution time: 29.9675 ms

Minimum execution time: 0.0002 ms

Mean execution time: 1.0828462520061568 ms

Execution time standard deviation: 0.2624776010196432 ms

Method: Telemetry/Proc/12/OptimizationProblemExecutionPerformance

Call count: 14800996

Maximum execution time: 29.5493 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.40387934431575606 ms

Execution time standard deviation: 0.06956059193751638 ms

Method: Telemetry/Proc/12/OptimizationProblemExecutionPerformance

Call count: 14438826

Maximum execution time: 16.1488 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.37022088212020476 ms

Execution time standard deviation: 0.12156739348265842 ms

Method: Telemetry/Proc/13/OptimizationProblemExecutionPerformance

Call count: 15857070
Maximum execution time: 20.217 ms
Minimum execution time: 0.0002 ms
Mean execution time: 1.0891627310847622 ms
Execution time standard deviation: 0.25677031620726753 ms

Method: Telemetry/Proc/13/OptimizationProblemExecutionPerformance

Call count: 14842831
Maximum execution time: 11.4656 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.4072256595052552 ms
Execution time standard deviation: 0.06667574369998125 ms

Method: Telemetry/Proc/13/OptimizationProblemExecutionPerformance

Call count: 14500117
Maximum execution time: 26.1624 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.375661576344538 ms
Execution time standard deviation: 0.11957347265205083 ms

Method: Telemetry/Proc/14/OptimizationProblemExecutionPerformance

Call count: 15857070
Maximum execution time: 24.2987 ms
Minimum execution time: 0.0002 ms
Mean execution time: 1.0956082442028203 ms
Execution time standard deviation: 0.2527766190751965 ms

Method: Telemetry/Proc/14/OptimizationProblemExecutionPerformance

Call count: 14859290
Maximum execution time: 21.5246 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.40709579175716987 ms
Execution time standard deviation: 0.06569317879986111 ms

Method: Telemetry/Proc/14/OptimizationProblemExecutionPerformance

Call count: 14520164
Maximum execution time: 17.4597 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.37531118753198733 ms
Execution time standard deviation: 0.11849743168393906 ms

Method: Telemetry/Proc/15/OptimizationProblemExecutionPerformance

Call count: 15857070
Maximum execution time: 26.2824 ms
Minimum execution time: 0.0002 ms
Mean execution time: 1.0885731187918668 ms
Execution time standard deviation: 0.26001015889159057 ms

Method: Telemetry/Proc/15/OptimizationProblemExecutionPerformance

Call count: 14833491
Maximum execution time: 30.7974 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.40568345974654424 ms
Execution time standard deviation: 0.06876451561357039 ms

Method: Telemetry/Proc/15/OptimizationProblemExecutionPerformance

Call count: 14488248
Maximum execution time: 18.6516 ms
Minimum execution time: 0.0002 ms

Mean execution time: 0.3743213143921914 ms
Execution time standard deviation: 0.1191220786834119 ms

Method: Telemetry/Proc/16/OptimizationProblemExecutionPerformance
Call count: 15857070
Maximum execution time: 23.6179 ms
Minimum execution time: 0.0002 ms
Mean execution time: 1.0934199534655367 ms
Execution time standard deviation: 0.25433957283254516 ms

Method: Telemetry/Proc/16/OptimizationProblemExecutionPerformance
Call count: 14845999
Maximum execution time: 15.2851 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.40723975484573 ms
Execution time standard deviation: 0.06420978835809833 ms

Method: Telemetry/Proc/16/OptimizationProblemExecutionPerformance
Call count: 14513218
Maximum execution time: 19.6328 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.3762945317985756 ms
Execution time standard deviation: 0.118392305431212 ms

Method: Telemetry/Proc/17/OptimizationProblemExecutionPerformance
Call count: 15857070
Maximum execution time: 28.8087 ms
Minimum execution time: 0.0002 ms
Mean execution time: 1.1077405725708014 ms
Execution time standard deviation: 0.2560590733741124 ms

Method: Telemetry/Proc/17/OptimizationProblemExecutionPerformance
Call count: 14838660
Maximum execution time: 23.0695 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.4122449741822735 ms
Execution time standard deviation: 0.06553317738015052 ms

Method: Telemetry/Proc/17/OptimizationProblemExecutionPerformance
Call count: 14488758
Maximum execution time: 25.2851 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.381233391681978 ms
Execution time standard deviation: 0.11814056390126626 ms

Method: Telemetry/Proc/18/OptimizationProblemExecutionPerformance
Call count: 15857070
Maximum execution time: 23.7972 ms
Minimum execution time: 0.0002 ms
Mean execution time: 1.091922206939714 ms
Execution time standard deviation: 0.2562963225775883 ms

Method: Telemetry/Proc/18/OptimizationProblemExecutionPerformance
Call count: 14843633
Maximum execution time: 16.314 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.406303671102599 ms
Execution time standard deviation: 0.06648287650777124 ms

Method: Telemetry/Proc/18/OptimizationProblemExecutionPerformance
Call count: 14498869
Maximum execution time: 17.9204 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.37652714191703607 ms
Execution time standard deviation: 0.11781697379256512 ms

Method: Telemetry/Proc/19/OptimizationProblemExecutionPerformance
Call count: 15857070
Maximum execution time: 30.8213 ms
Minimum execution time: 0.0002 ms
Mean execution time: 1.0899489842765733 ms
Execution time standard deviation: 0.2622750483422762 ms

Method: Telemetry/Proc/19/OptimizationProblemExecutionPerformance
Call count: 14801513
Maximum execution time: 25.2769 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.40488495339634706 ms
Execution time standard deviation: 0.06850700669745975 ms

Method: Telemetry/Proc/19/OptimizationProblemExecutionPerformance
Call count: 14440614
Maximum execution time: 18.517 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.3736298215574226 ms
Execution time standard deviation: 0.12106001462251648 ms

Method: Telemetry/Proc/20/OptimizationProblemExecutionPerformance
Call count: 15857070
Maximum execution time: 25.2303 ms
Minimum execution time: 0.0002 ms
Mean execution time: 1.088636699484834 ms
Execution time standard deviation: 0.25722294679325497 ms

Method: Telemetry/Proc/20/OptimizationProblemExecutionPerformance
Call count: 14867399
Maximum execution time: 25.5308 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.40473169477050897 ms
Execution time standard deviation: 0.06832531588433993 ms

Method: Telemetry/Proc/20/OptimizationProblemExecutionPerformance
Call count: 14546033
Maximum execution time: 14.6691 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.375044770811364 ms
Execution time standard deviation: 0.11847936700194522 ms

Method: Telemetry/Proc/21/OptimizationProblemExecutionPerformance
Call count: 15857070
Maximum execution time: 23.6263 ms
Minimum execution time: 0.0002 ms
Mean execution time: 1.0883652042526932 ms
Execution time standard deviation: 0.25876545736178796 ms

Method: Telemetry/Proc/21/OptimizationProblemExecutionPerformance
Call count: 14811763
Maximum execution time: 15.2567 ms

Minimum execution time: 0.0002 ms
Mean execution time: 0.4051173227454467 ms
Execution time standard deviation: 0.06648229410242834 ms

Method: Telemetry/Proc/21/OptimizationProblemExecutionPerformance
Call count: 14469309
Maximum execution time: 13.7629 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.37348156574026814 ms
Execution time standard deviation: 0.12010090570659737 ms

Method: Telemetry/Proc/22/OptimizationProblemExecutionPerformance
Call count: 15857070
Maximum execution time: 22.2966 ms
Minimum execution time: 0.0002 ms
Mean execution time: 1.0879653685074775 ms
Execution time standard deviation: 0.2576012181634194 ms

Method: Telemetry/Proc/22/OptimizationProblemExecutionPerformance
Call count: 14822253
Maximum execution time: 24.0172 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.40596488966281413 ms
Execution time standard deviation: 0.06642210473219823 ms

Method: Telemetry/Proc/22/OptimizationProblemExecutionPerformance
Call count: 14479071
Maximum execution time: 15.4626 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.3764990789809571 ms
Execution time standard deviation: 0.11919650244555714 ms

Method: Telemetry/Proc/23/OptimizationProblemExecutionPerformance
Call count: 15857070
Maximum execution time: 24.5201 ms
Minimum execution time: 0.0002 ms
Mean execution time: 1.0842693774574605 ms
Execution time standard deviation: 0.25778998135954373 ms

Method: Telemetry/Proc/23/OptimizationProblemExecutionPerformance
Call count: 14862907
Maximum execution time: 24.4711 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.4029442837999986 ms
Execution time standard deviation: 0.06808058313588036 ms

Method: Telemetry/Proc/23/OptimizationProblemExecutionPerformance
Call count: 14516120
Maximum execution time: 14.7656 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.3732159120894485 ms
Execution time standard deviation: 0.1173209956159693 ms

Method: Telemetry/Proc/24/OptimizationProblemExecutionPerformance
Call count: 15857070
Maximum execution time: 27.3942 ms
Minimum execution time: 0.0002 ms
Mean execution time: 1.0967293663268811 ms
Execution time standard deviation: 0.260959984768826 ms

Method: Telemetry/Proc/24/OptimizationProblemExecutionPerformance

Call count: 14852262

Maximum execution time: 24.8881 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.4092484509497304 ms

Execution time standard deviation: 0.06950058470818833 ms

Method: Telemetry/Proc/24/OptimizationProblemExecutionPerformance

Call count: 14518620

Maximum execution time: 15.8521 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.37773903081005694 ms

Execution time standard deviation: 0.11904334954455702 ms

Method: Telemetry/Proc/25/OptimizationProblemExecutionPerformance

Call count: 15857070

Maximum execution time: 38.1418 ms

Minimum execution time: 0.0002 ms

Mean execution time: 1.094690120961666 ms

Execution time standard deviation: 0.2518654269350071 ms

Method: Telemetry/Proc/25/OptimizationProblemExecutionPerformance

Call count: 14890568

Maximum execution time: 16.4598 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.40672272248442 ms

Execution time standard deviation: 0.06528136302795395 ms

Method: Telemetry/Proc/25/OptimizationProblemExecutionPerformance

Call count: 14562922

Maximum execution time: 19.4409 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.376525841057243 ms

Execution time standard deviation: 0.1170218165933666 ms

Method: Telemetry/Proc/26/OptimizationProblemExecutionPerformance

Call count: 15857070

Maximum execution time: 29.9081 ms

Minimum execution time: 0.0002 ms

Mean execution time: 1.0971012370443978 ms

Execution time standard deviation: 0.25096567646721785 ms

Method: Telemetry/Proc/26/OptimizationProblemExecutionPerformance

Call count: 14903834

Maximum execution time: 20.1928 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.40706869998684886 ms

Execution time standard deviation: 0.06493580328666963 ms

Method: Telemetry/Proc/26/OptimizationProblemExecutionPerformance

Call count: 14578514

Maximum execution time: 17.5556 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.37502934970601987 ms

Execution time standard deviation: 0.11724642091080187 ms

Method: Telemetry/Proc/27/OptimizationProblemExecutionPerformance

Call count: 15857070

Maximum execution time: 26.1868 ms
Minimum execution time: 0.0002 ms
Mean execution time: 1.0965487830725877 ms
Execution time standard deviation: 0.251526709331354 ms

Method: Telemetry/Proc/27/OptimizationProblemExecutionPerformance

Call count: 14872414
Maximum execution time: 22.4531 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.4078356138082393 ms
Execution time standard deviation: 0.0645212424867144 ms

Method: Telemetry/Proc/27/OptimizationProblemExecutionPerformance

Call count: 14552662
Maximum execution time: 17.7766 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.37663234201411544 ms
Execution time standard deviation: 0.11787122376816017 ms

Method: Telemetry/Proc/28/OptimizationProblemExecutionPerformance

Call count: 15857070
Maximum execution time: 25.4004 ms
Minimum execution time: 0.0002 ms
Mean execution time: 1.0987801429078192 ms
Execution time standard deviation: 0.25337556550763607 ms

Method: Telemetry/Proc/28/OptimizationProblemExecutionPerformance

Call count: 14866717
Maximum execution time: 15.384 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.4084733670318538 ms
Execution time standard deviation: 0.06394551032133079 ms

Method: Telemetry/Proc/28/OptimizationProblemExecutionPerformance

Call count: 14532772
Maximum execution time: 14.4623 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.37679966154421135 ms
Execution time standard deviation: 0.1173679014844708 ms

Method: Telemetry/Proc/29/OptimizationProblemExecutionPerformance

Call count: 15857070
Maximum execution time: 27.9496 ms
Minimum execution time: 0.0002 ms
Mean execution time: 1.0883401324770736 ms
Execution time standard deviation: 0.2576644008358268 ms

Method: Telemetry/Proc/29/OptimizationProblemExecutionPerformance

Call count: 14830702
Maximum execution time: 19.1145 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.40625316902060826 ms
Execution time standard deviation: 0.06646180396590534 ms

Method: Telemetry/Proc/29/OptimizationProblemExecutionPerformance

Call count: 14468504
Maximum execution time: 15.9604 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.37494224016528604 ms

Execution time standard deviation: 0.11892011929152276 ms

Method: Telemetry/Proc/30/OptimizationProblemExecutionPerformance

Call count: 15857070

Maximum execution time: 28.9653 ms

Minimum execution time: 0.0002 ms

Mean execution time: 1.0956874372565957 ms

Execution time standard deviation: 0.2552248263241412 ms

Method: Telemetry/Proc/30/OptimizationProblemExecutionPerformance

Call count: 14844417

Maximum execution time: 16.4928 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.40675655415777096 ms

Execution time standard deviation: 0.06436325693929383 ms

Method: Telemetry/Proc/30/OptimizationProblemExecutionPerformance

Call count: 14508747

Maximum execution time: 35.8206 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.37584259563558853 ms

Execution time standard deviation: 0.1185320908514447 ms

Method: Telemetry/Proc/31/OptimizationProblemExecutionPerformance

Call count: 15857070

Maximum execution time: 23.0499 ms

Minimum execution time: 0.0002 ms

Mean execution time: 1.094104432079902 ms

Execution time standard deviation: 0.26307325010582494 ms

Method: Telemetry/Proc/31/OptimizationProblemExecutionPerformance

Call count: 14853464

Maximum execution time: 18.1552 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.406469034199794 ms

Execution time standard deviation: 0.06951403750143612 ms

Method: Telemetry/Proc/31/OptimizationProblemExecutionPerformance

Call count: 14512133

Maximum execution time: 15.1967 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.37641935667209825 ms

Execution time standard deviation: 0.1193859032222972 ms